

2
Book

Science TOPICS

FOR SENIOR STUDENTS

PETER GRIBBEN AND
WENDY ST GERMAIN



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FOR SENIOR STUDENTS

BOOK

2

**PETER GRIBBEN AND
WENDY ST GERMAIN**



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The three *Science for Senior Students* books present a wide variety of material aimed at students studying science subjects in Years 11 and 12. Many of the work sheets can be used as extension work for more able Year 9 and 10 students. The activities are designed to gain the interest of students and extend their knowledge, understanding and recall of science, something not always encouraged by conventional text books.

Over the three books in this series many branches of science are represented, including astronomy, biology, chemistry, ecology, geology and physics.

The broad range of subject areas is covered in a variety of styles to suit most ability levels. These include comprehension exercises, assignments, tests, graphing exercises, and calculations. The majority of topics are covered in a single page, with none being more than two pages.

The design highlights definitions, and key concepts, separate from the main text.

Many of the sheets can be used by teachers who have little prior experience in science.

In order to extend self-motivated students, and improve their fact-finding skills, most topics include a 'research' section. Outline answers to these are provided in the Answers section.

The worksheets in *Science for Senior Students* are intended to:

- provide relevant material for class use to cover the absence of the regular teacher
- introduce new work in an easily assimilated manner
- provide homework or extension exercises
- review and revise specific topics prior to tests/examinations
- cover the 'gaps' left by standard text books.



Where calculations or graphing work are included on a work sheet, these are indicated by the use of an icon graphic.



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Scientific Method I

The word 'Science' comes from the Latin '*scientia*', meaning '*to know*'. It is used to describe organised knowledge. A scientist could be described as someone who studies a particular area of knowledge in a methodical way. There are many areas of science, including astronomy, biology, chemistry, geology, meteorology, paleontology and physics.

Scientists deal with problems in a certain way. They may notice something interesting and then try to explain why it happens. The main way they do this is to gather more information (often by carrying out experiments) and then make an educated guess. If this doesn't work, they think again and perhaps try different experiments. The following terms are often used to describe scientific method.

- **Observation:** seeing, hearing, touching, smelling or tasting.
- **Inference:** a judgement based on an observation.
- **Hypothesis:** a possible explanation based on observation.
- **Experiment:** to do something to test a hypothesis.
- **Results:** observations/measurements from an experiment.
- **Conclusion:** a final decision reached after doing experiments.

Observations involve more than seeing — they can also involve hearing, smelling, touching and tasting. Observations are *quantitative* when numbers are given (for example, 25°C) and *qualitative* when they involve descriptions (for example, 'hot', 'cold'). An hypothesis should be written as a statement which can be proved or disproved by experiment.

Read through the following statements and decide which are hypotheses and which are observations.

- 1 Food additives can cause allergic reactions in some people.
- 2 A packet of cold relief tablets contained paracetamol, Vitamin C, glucose and colouring.
- 3 Paper packaging is less harmful to the environment than plastic.
- 4 Brown bread is better for you than white bread.
- 5 White cars are very common.

Demelza and Jess decided to study the ecology of a local park for their science project. Every day after school they went to the park and made some observations of living things. Each Friday they made a summary of their results. They did this for a term:

Week	1	2	3	4	5	6	7	8	9	10
Flowers	none	none	a few	a few	lots	many	many	very many	very many	very many
Bees	none	none	none	20	25	35	50	60	65	80
Butterflies	6	3	7	8	12	12	20	25	24	30
Birds	4	5	3	5	6	9	9	14	18	17
Beetles	12	11	13	10	8	6	5	4	5	4

- 6 Are the data recorded for the flowers quantitative or qualitative? Explain your answer.
- 7 Between which two weeks did the number of bees increase the most?
- 8 Demelza said to Jess, 'We should include in our conclusion that the number of birds depends upon the number of butterflies'. Discuss this statement.



RESEARCH

- 9 Provide a brief definition of 'paleontology'.

Scientific Method II

In science experiments it is important to change only one thing (or *variable*) at a time.

In Australia we expend a lot of effort keeping hot drinks and food hot, and cold items cold. It is therefore important to know the insulating properties of materials. A teacher gave a class a practical assignment to investigate the rate at which water cooled in containers of different colours. Each set of students was given a vacuum flask of water at 80°C and a coloured aluminium beaker (either silver, white, red or black). The beakers were identical apart from their colours. The students were allowed access to the normal range of school laboratory equipment.

The table below shows how the students in one group selected and used their equipment. One essential piece of equipment is missing.

1 State what the missing piece of apparatus is.

2 Why is the item in Question 1 used?

One item could be replaced by a more appropriate piece of equipment.

3 Name the item that should be replaced.

4 Name the replacement.

5 Explain your answers to Questions 3 and 4.

6 Name two factors which would need to be controlled during this experiment.

Item	Why it was used
Vacuum flask	To store hot water
Aluminium beaker	To hold water during experiment
Graph paper	To plot graph of temperature against time
Stopwatch	To take readings of time
250 mL glass beaker	To measure volume of water

There are often two types of variables in an experiment, *independent* variables and *dependent* variables. The person carrying out the experiment decides on what the independent variable is. The dependent variable depends on, or changes with, the independent variable.

The students decided to measure the temperature of the water every minute. A sample set of results is shown, on the graph below.

7 Which is the independent variable in this experiment?

8 Which is the dependent variable in this experiment?

9 Explain your answers to Questions 7 and 8.

10 What was the initial temperature of the water?

11 What was the temperature of the room?

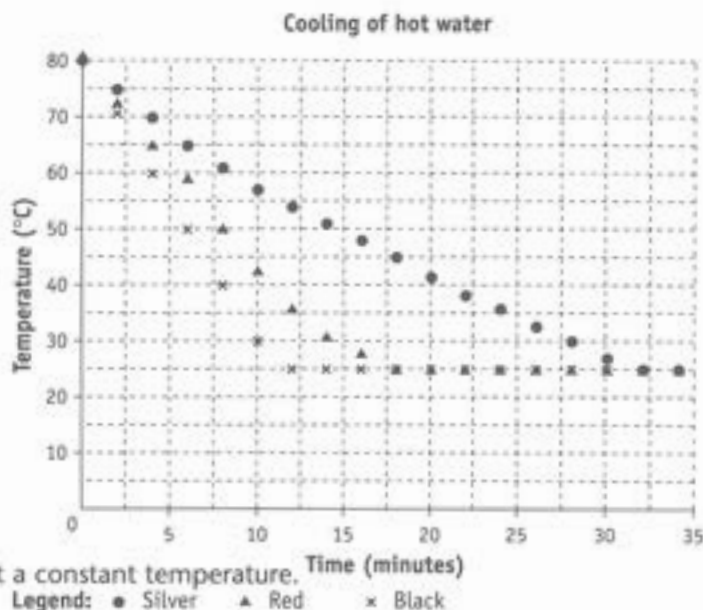
12 Which set of results is missing?

13 A 'conclusion' is what you decide to be true after you have carried out an experiment. It is put at the end of the practical write-up. Write a conclusion for the experiment shown here.



RESEARCH

14 Explain how a vacuum flask keeps its contents at a constant temperature.



Tables

In science we often use tables to present information. Tables can present a lot of information in a small space. Most tables are in the form of boxes divided into *rows* and *columns*. Rows are horizontal, going across (→) a table from left to right. Columns are vertical, going up and down a table.(↑↓).

REFRIGERATORS

This table shows the details of some refrigerators or 'fridges'. It is divided into six columns and seven rows. Study the table carefully and answer the questions.

Brand	Capacity (litres)	Automatic defrost	Choice of colour	Left/right door	Cost (\$)
Koola	300	✓	X	✓	750
Technocool	450	✓	X	X	900
Freezefast	410	X	X	✓	650
Polar	370	X	X	X	625
Arctic	400	✓	✓	✓	995
Supercool	310	✓	✓	X	595

- Which is the cheapest fridge?
- Which fridge has the largest capacity?
- Which is the most expensive fridge?
- Which fridge has the second-largest capacity?
- Which fridges can be bought in a colour other than white?
- Which fridge has the most additional features?
- Work out the cost per litre of capacity for each fridge.
- Name *two* other pieces of information about the fridges that you might need to know before deciding which one you might want to buy.

TELEVISIONS

The table shows some information about TVs for sale in the local 'TellyWorld' shop.

Study the table and then answer the questions.

Model	Screen size (cm)	Voltage	Price (\$)	Guarantee (months)	Made in
GTV-34	34	mains	340	12	Taiwan
Newtel	60	mains	895	12	Korea
Digitv	34	mains	410	24	Malaysia
Techtv	54	mains	850	18	Japan
Supatv	34	12 V/mains	460	12	Indonesia
Dragon	34	mains	330	12	China

- How many types of TV are there?
- How many different screen sizes are there?
- Who makes the biggest TV screen?
- What is the most common screen size?
- Which TV has the longest guarantee?
- Another shop is selling the Tech tv at \$30 less than TellyWorld's advertised price. How much would you pay at this shop?
- Which TV could be used with a car battery?
- Would you buy a TV, based only on the information given in an advertisement? Explain your answer.



RESEARCH

- Explain, with the aid of a diagram, how screen sizes are measured.



The Pendulum: Assignment

NAME _____ DUE DATE _____

This sheet contains practical instructions and sample results for a science experiment. Read the sheet and answer the questions.

A simple pendulum consists of a bob (mass) suspended from a string, which is attached to a fixed object. The *periodic time* of a pendulum is the time it takes for the mass to swing back and forth, that is from A to B and back to A. [See Diagram (i)]. In this experiment the bob is a 50 g mass hanger, to which extra masses can be attached.

The aim of the experiment is to determine whether there is a relationship between periodic time and:

- A the mass of the bob
- B the length of the pendulum
- C the distance the bob is pulled from vertical

PART A: 'MASS OF THE BOB'

- Set the string length to 40 cm [see Diagram (ii)].
- Pull the bob back so that it is 15 cm from the vertical. [see Diagram (iii)]. Let the bob go.
- Measure the time taken for 20 periods (swings) with just the 50 g mass carrier attached.
- Enter the results in the table below.
- Repeat the measurement twice more, then calculate an average for the three tests.
- Use the averaged figure to calculate the time of one period. (Round to two decimal places.)
- Repeat this procedure for total masses of 100 g and 150 g.

Results Part A:

Mass of bob (grams)	Time for 20 periods (seconds)				Periodic time (seconds)
	Test 1	Test 2	Test 3	Average	
50	12.63	12.55	12.62		
100	12.48	12.73	12.27		
150	12.61	12.62	12.65		

- 1 Complete the table.
- 2 On a separate piece of graph paper, plot mass (x-axis) against periodic time (y-axis).
- 3 What effect does changing the mass of the bob have on the periodic time? Explain your answer.

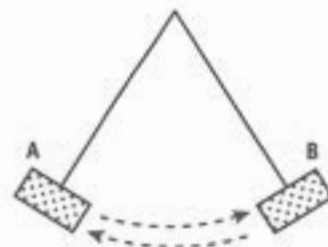


Diagram (i)

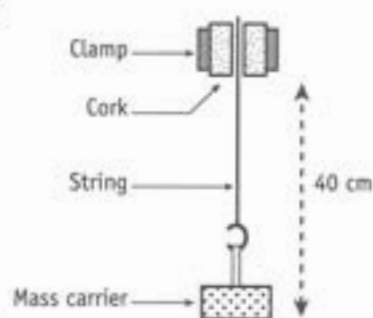


Diagram (ii)

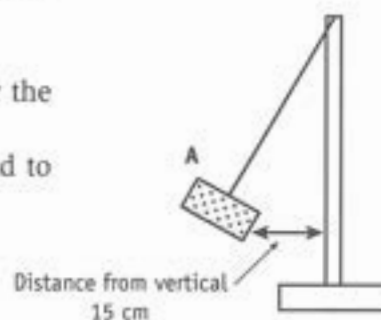


Diagram (iii)



The Pendulum: Assignment

NAME _____ DUE DATE _____

PART B: 'LENGTH OF PENDULUM'

- Set the pendulum length to 50 cm (using a 50 g mass carrier). Keep the same distance from vertical as was used in Part A.
- Measure the time for 20 periods, and record your results in the table below.
- Repeat the measurement twice more, then calculate the average time for 20 periods.
- Use the average time to calculate the time for one period (to two decimal places).
- Repeat the procedure for lengths of 75 cm and 100 cm.



Results Part B:

Pendulum length (centimetres)	Time for 20 periods (seconds)				Periodic time (seconds)
	Test 1	Test 2	Test 3	Average	
50	14.16	14.22	14.25		
75	17.53	17.44	17.50		
100	20.32	20.09	20.25		

- Complete the table.
- On a separate piece of graph paper, plot mass (x-axis) against periodic time (y-axis).
- What effect does changing the pendulum length have on the periodic time? Explain your answer.

PART C: 'DISTANCE FROM VERTICAL'

- Set the pendulum length to 50 cm (using a 50 g mass carrier). Set the distance from vertical to 10 cm.
- Measure the time for 20 periods and record your results in the table below.
- Repeat the measurement twice more, then calculate the average time for 20 periods.
- Use the average time to calculate the time for one period (to two decimal places).
- Repeat the procedure for a distance from the vertical of 20 cm.

Results Part C

Distance (centimetres)	Time for 20 periods (seconds)				Periodic time (seconds)
	Test 1	Test 2	Test 3	Average	
10	14.38	14.50	14.49		
20	14.47	14.34	14.61		

- Complete the table.
- On a separate piece of graph paper, plot mass (x-axis) against periodic time (y-axis).
- What effect does changing the distance from vertical have on the periodic time? Explain your answer.



Surveys

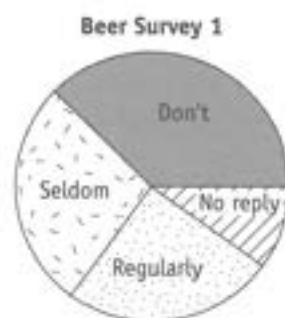
A science class carried out a survey of pupils. One of the things the survey involved was counting the number of pupils in each form. There were 23 pupils in S1, 18 in S2, 13 in S3, 17 in S4, 9 in S5 and 24 in S6.

- 1 How many pupils were there in total?

At the same time each pupil was asked to name his or her favourite sport. Twelve said basketball, 19 Australian football, 15 netball, 3 rugby union, 8 rugby league, 3 golf, 17 soccer, 3 lawn bowls, 14 fishing, 2 roller hockey, 4 canoeing and 2 baseball.

- 2 Convert to a percentage of all the pupils the number of pupils for each sport.
- 3 Present this information in the form of a labelled column graph. Give the graph a title.
- 4 What was the third most popular sport?
- 5 How many people did not respond to the survey?

A survey was commissioned to investigate the beer-drinking habits of adults in a particular community. Questions were asked of 120 people on a Saturday morning in a busy shopping centre. The first question was 'How often do you drink beer?' The responses to choose from were: 'Don't', 'Seldom' or 'Regularly'. Questions 6 and 7 refer to the 'Beer Survey' chart. You will need a protractor.



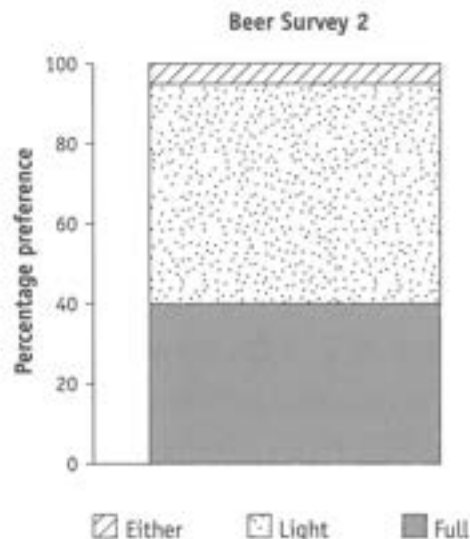
- 6 Which category in the pie chart (sector graph) was not mentioned in the given list of responses?
- 7 Calculate the number of responses in each category and put these results in table form.

Those people who said they drank beer on a regular basis were asked a second question, 'Do you drink full-strength beer or light beer?' Their responses are shown in the 'Beer Survey 2' chart. Questions 8, 9 and 10 refer to this chart.

- 8 What percentage of drinkers said they preferred light beer?
- 9 What percentage preferred full-strength beer?
- 10 What percentage drank either?

When the sales figures from local bottle shops and other outlets were studied, it was found that approximately four times as much full-strength beer was sold as light beer.

- 11 Give three possible explanations as to why these figures do not agree with those in 'Beer Survey 2'.
- 12 Give examples of three groups who might want this type of survey done.
- 13 Give three ways in which the accuracy of this survey could be improved.



A newspaper had a phone-in line where readers rang one number if they wanted to answer 'yes' to a particular question and another if they wanted to vote 'no'.

- 14 Comment on the advantages and disadvantages of this system.



RESEARCH

- 15 You have been given an assignment to find out the television viewing habits of a representative sample of males and females of your age group. Describe how you would carry out this task. Include how you would ensure that this was a fair and accurate survey.
- 16 Find out how commercial organisations carry out opinion polls to find out how the general public feels about a particular issue. Comment on the accuracy of these polls.



Plant Growth

A science class was given a research project to investigate the effect that Gro-betta, a liquid fertiliser, had on the growth of plants. The students needed to design and then carry out an experiment to see what effect, if any, this product had on the growth of plants when compared to just watering them.

One student submitted a report which is shown below. Read it carefully and then answer the questions that follow.

Title: Using Gro-betta

Aim: To see how tall plants will grow.

Hypothesis: When Gro-betta is added to plants they will grow taller.

Method:

- (i) I got 6 plant pots and filled them up with soil.
- (ii) Into each one I put a plant seedling.
- (iii) I labelled each pot as 'A', 'B', 'C', etc.
- (iv) Every few days I watered them with 250 mL of fertiliser diluted with water.
- (v) I measured the height of the plants just after I watered them.

Results:

Plant	Height of plant (cm)								
	Day 1	Day 5	Day 8	Day 14	Day 19	Day 26	Day 30	Day 33	Day 38
A	12.2	12.9	13.8	14.8	15.4	15.6	16.2	16.7	17.1
B	12.1	12.6	12.7	12.8	12.6	12.5	12.4	12.4	12.4
C	12.3	12.9	13.7	14.7	15.6	16.2	17.0	17.3	17.7
D	12.4	12.7	13.1	13.9	14.9	15.7	16.7	18.0	18.5
E	11.9	12.5	13.5	14.3	15.3	16.1	16.9	17.4	17.9
F	13.6	14.2	15.5	16.2	17.1	17.9	18.5	19.0	19.6

Conclusion: My results show that all plants grow taller when Gro-betta is added. Plant B didn't work, because it died.

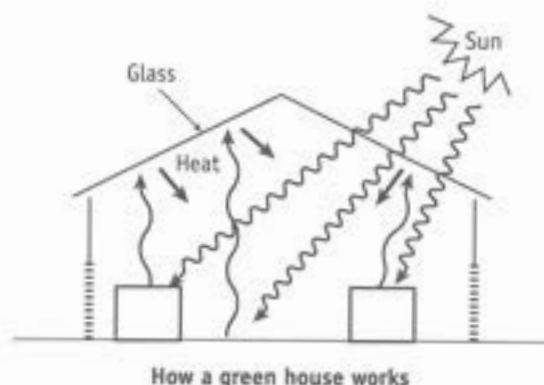
- 1 What was the average increase in height over the period of the experiment? (Show your working.)
- 2 Look at the stated aim. Comment on whether or not this was what the group was asked to investigate.
- 3 Study parts (i) and (ii) of the method. For both parts state two extra details which should be included to make it a fair test.
- 4 Is this experiment a fair test on how well Gro-betta helps plants grow? Explain your answer.
- 5 On graph paper, show how Plant A grew. Indicate on your graph which is the 'dependent' and which is the 'independent' variable.
- 6 What height do you think Plant A would be one week after the last measurement was made? Explain how you arrived at this answer, including what assumptions you made.

The Atmosphere and the Greenhouse Effect

Greenhouses are used to grow plants that would not thrive in outdoor conditions. In a greenhouse, the walls are made of a transparent material, such as glass. The glass allows solar radiation (sunlight) through. Objects inside are warmed up and this energy is then given off as *infra-red* or heat radiation which has a different wavelength compared to the original radiation. The glass does not let this through and, because of this, the heat is trapped inside. The interior temperature remains higher than outside.

The atmosphere surrounds the Earth. It is made up of gases, water vapour and airborne particles held near the Earth by the force of gravity. As we go higher into the atmosphere, the amount of air becomes less and less, and air pressure drops. This is why mountain climbers may need to take oxygen with them, and why high-flying aircraft need pressurised cabins. High-flying planes pump in extra air to keep the pressure inside up to normal levels. Most of the mass (80%) of the atmosphere lies within 17 km of sea level. However, traces of atmosphere can be found thousands of kilometres from the Earth's surface. The atmosphere acts like the glass of the

greenhouse because it contains the so-called *greenhouse gases*. These gases include water vapour (H_2O), carbon dioxide (CO_2), methane (CH_4), nitrous oxide (NO), ozone (O_3) and chlorofluorocarbons.



The atmosphere is transparent to short-wave solar radiation. Sunlight passes through the atmosphere where it reaches the Earth's surface and heats it up. Then, because it is hot, the Earth re-emits some warmth, in the form of infra-red radiation, back into space. Some of this radiation is absorbed by the greenhouse gases and kept in the atmosphere. (You may have noticed that a cloudy night is warmer than a clear one.) The air warms up and the temperature of the Earth is maintained. Because of the greenhouse effect, life as we know it is possible on Earth. Without it, the temperature of the planet would vary tremendously between either being too hot or too cold to allow us to live.

A current problem is that some human activities are upsetting the balance of gases in the atmosphere. They are causing an increase in the amount of greenhouse gases and so the air retains more warmth. Some examples of these activities are:

- The increase in the production of carbon dioxide by the burning of *fossil fuels* such as coal, oil and gas. Carbon dioxide contributes to about half the total warming effect. Estimates vary, but carbon dioxide has been increasing in the atmosphere by around 0.4% annually.
- The increase in the production of methane, mainly from industrial activities involving the use of natural gas, which is mainly methane. (Animals such as cows produce large quantities of methane naturally as part of their digestive processes.)
- Chlorofluorocarbons — also known as CFCs — have been used as refrigerants, solvents in dry-cleaning and propellants for aerosol cans. They destroy the *ozone layer*, which helps protect us from the harmful effects of ultra-violet rays.

Some scientists predict that the Earth will become warmer and this will cause major changes in our climate and weather patterns. Rises of between 2°C to 7°C in average global temperature have been forecast over this century.

The Atmosphere and the Greenhouse Effect

Agriculture could be affected — some areas will become more productive, others will suffer if temperatures move up. It is very difficult to forecast exactly what will happen.

There has been a rise in sea level of about 1.5 mm per year over the past few decades. This rise is probably due to the melting of polar ice caps due to the increased level of the greenhouse effect. Many low-lying islands in the Pacific Ocean risk being submerged if warming causes the sea-level to rise. Islands threatened include the Maldives, Fiji and Vanuatu.

What steps can we take to make sure that the greenhouse effect does not keep increasing?

We can try to find alternative sources of energy that do not involve the generation of carbon dioxide. Trees (and all green plants) take carbon dioxide out of the atmosphere as part of their process of photosynthesis — the means by which they make their food. So, planting more trees could be of some assistance in keeping the levels of carbon dioxide at current levels. When trees die, or deforestation occurs, carbon dioxide is released.

- 1 Draw a labelled diagram showing how sunlight reaches the Earth and how this energy is prevented from escaping into space. (Hint: the greenhouse diagram provided earlier may help.)
- 2 Give the names and chemical formulae of three greenhouse gases.
- 3 What does 'transparent' mean?
- 4 Would we be able to live on Earth if there was no greenhouse effect? Give an explanation for your answer.
- 5 Why shouldn't you leave a dog in a car on a sunny day?
- 6 What is the source of energy for all life on Earth?
- 7 If the planet becomes warmer, what might happen to the polar ice caps? What types of countries might be affected by this?
- 8 What is another name for 'infra-red'?
- 9 Many aerosol (spray) cans are advertised as *CFC-free* or *ozone-friendly*. Why might these be better than a product containing chlorofluorocarbons?
- 10 Suggest two ways in which we can keep levels of carbon dioxide steady in the world.
- 11 What is another name for 'solar radiation'?
- 12 Why might an aircraft flying at 10 000 m need a pressurised cabin?
- 13 How might agriculture be affected if we have a permanent increase in average temperatures?
- 14 What affect will deforestation have on carbon dioxide levels?



RESEARCH

- 15 Name the three most common gases present in the atmosphere, and their relative proportions.
- 16 Give definitions for: 'troposphere', 'stratosphere', 'mesosphere' and 'ionosphere'.
- 17 Find out what chemicals are being used to replace CFCs. State whether these substances themselves have potential to harm the environment.
- 18 Methane can be produced by organisms other than animals. Name one of these sources of methane.

Pollution

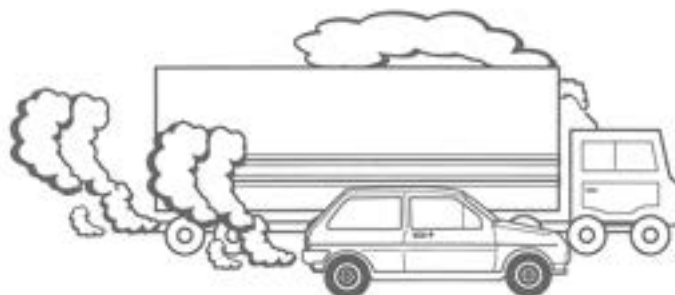
A substance *pollutes* when its presence in air, water or soil harms living things.

The main cause of air pollution is the burning of fossil fuels, such as coal and oil. This happens in factories, power stations, houses and the engines of cars, buses, and aeroplanes. Poisonous gases from chimneys are quickly spread out in the air but, if there is not too much of them, they can change into less harmful substances over time.

The smoke produced by burning contains tiny particles of dust, which are mainly carbon (C). This dust settles on plant leaves, cutting out some of the light taken in by the plants and so reducing photosynthesis. People also breathe in the dust and this can worsen illnesses such as asthma and bronchitis.

Sulfur dioxide (SO_2) is a gas produced by burning fossil fuels. It can dissolve in water vapour, forming an acid, which falls as *acid rain*. As well as damaging buildings, plants and wildlife this can affect people's lungs.

Petrol and diesel (internal combustion) engines, especially when they are not running properly, produce fumes which gather in city streets. As well as sulfur dioxide, carbon monoxide (CO) and oxides of nitrogen are produced. Some petrol ('super') contains lead (Pb). Cars burning leaded petrol emit lead compounds. These can collect in the bodies of people living in areas where traffic is dense. Lead compounds can cause damage to health — especially that of the brain so Lead Replacement Petrol (LRP) is now being used. Lead was once used for water pipes. It is soft, easily worked and lasts for a long time. It has been replaced by copper (Cu) or polymers such as polyvinyl chloride (PVC)



Sometimes, in areas of heavy traffic and dense industry, fog and smoke can combine to form *smog* which is very harmful to lungs.

Many people are worried about the ozone layer. This is a region of the atmosphere from about 20 km to 50 km above the Earth's surface. Ozone (O_3) is a type of oxygen. It is poisonous if breathed in, but it protects us from the Sun's cancer-causing ultraviolet (UV) radiation. There is a group of chemicals called chlorofluorocarbons (often abbreviated to CFCs) which poses a threat to the ozone layer. They have been used in aerosols, refrigerators, air-conditioning units and polystyrene packaging. When they escape into the atmosphere they eventually react with and destroy ozone molecules. Because of this, the use of CFCs has been banned (or reduced) in many countries.

Most pollution is only harmful for a while and then becomes safe. One example is *sewage*. Sewage is liquids or waste matter carried off by sewers from houses and factories. If it is treated and then discharged in small quantities into the open sea, it can be decomposed and made harmless by some strains of bacteria able to decompose it.

Pollution

If too much is discharged too quickly, then harm can be done to the marine environment into which it is discharged. Also the bacteria involved can grow extremely quickly and use up all of the oxygen (Biological Oxygen Demand, BOD) and this can kill fish and other creatures living in the water.

Some substances remain poisonous for a very long time, for example, compounds containing mercury (Hg) and cyanide (CN⁻). These long lasting pollutants accumulate in fish and other aquatic creatures. They can then spread through food webs to water birds and, eventually, to people.

Other pollutants can result from modern farming methods. Many crops need applications of *pesticides* — substances which kill pests. These include sprays such as insecticides which kill insects that attack crops, and herbicides which kill weeds. These chemicals can be washed into rivers and ponds and spread through food webs into many organisms. Fertilisers, if used in excess, can also find their way into waterways and cause problems.

Crude oil (petroleum) is sometimes spilt into the sea by oil tankers or off-shore rigs. This oil can kill sea birds by poisoning them or by sticking their feathers together so that they cannot fly. When it is washed ashore, oil kills all forms of life on rocks or in mud and sand.

- 1 Draw up a table giving the chemical symbols and formulae for any elements or compounds mentioned in the text on these sheets.
- 2 Give two examples of 'fossil fuels'.
- 3 What is 'smog'?
- 4 What is 'ozone'? From what does it protect us?
- 5 Why is there a move to run cars on petrol that does not contain lead?
- 6 Give a definition of 'a pollutant'.
- 7 What is formed when sulfur dioxide in the atmosphere reacts with rain?
- 8 What does 'aquatic' mean?
- 9 What is a 'food web'?
- 10 What is 'sewage'?

You have just bought an old house. It has water pipes made out of lead. The plumber advises you to replace them although they are perfectly serviceable and not leaking.

- 11 Would you accept the advice? Explain your answer.
- 12 Why do farmers put fertilisers on their soil?
- 13 Why do farmers put pesticides on their crops?
- 14 What is the difference between a herbicide and an insecticide?



RESEARCH

- 15 Explain what causes *algal bloom* and describe the effects of this phenomenon.
- 16 An aerosol canister states on its label that the propellant (substance which pushes the contents out of the can) is *ozone-friendly*. Give the name of a propellant that is ozone friendly.
- 17 Minamata, a place in Japan, is associated with mercury poisoning. Describe what happened there and why it raised awareness of the problems associated with mercury.

Quadrat Sampling

A quadrat is a tool used for estimating the population of an organism. The quadrat is a square of known size, often made of wire. It is thrown at random and the area enclosed by it is studied. The grid below shows the population of three types of plant in an area measuring 6 metres by 6 metres, meaning each box is 1 m^2 . This exercise is intended to simulate random sampling and estimation of populations.

To decide where the (imaginary) quadrat lands, roll a die twice. The first throw will correspond to the letters (A to F) on the x-axis and the second throw the numbers (1 to 6) on the y-axis. So, if you roll 3 then 4, for example, then you will study box C4.

Roll the die to select a box. Count the separate types of plants present (see the key). Repeat this to sample ten 1 m^2 areas.

	A	B	C	D	E	F
1	    	    	  	   	    	   
2	   	    	   	    	   	    
3	    	    	    	    	    	    
4	    	    	   	   	    	    
5	    	    	  	   	    	    
6	    	    	    	    	    	    

Key:

○ Dandelion

▲ Daisy

★ Oxalis

- 1 Record your results in table form.
- 2 The grid only shows part of a field measuring 20 m by 50 m. Calculate how many of each plant you would expect to find in the total area. Show your working.
- 3 What assumption are you making in your answer to Question 2?
- 4 Why might you use quadrats to estimate a population, rather than counting everything?
- 5 Give one way of improving the accuracy of the method outlined above.
- 6 Compare your results with those of other members of the class. Comment on what you find.



Climate: Assignment

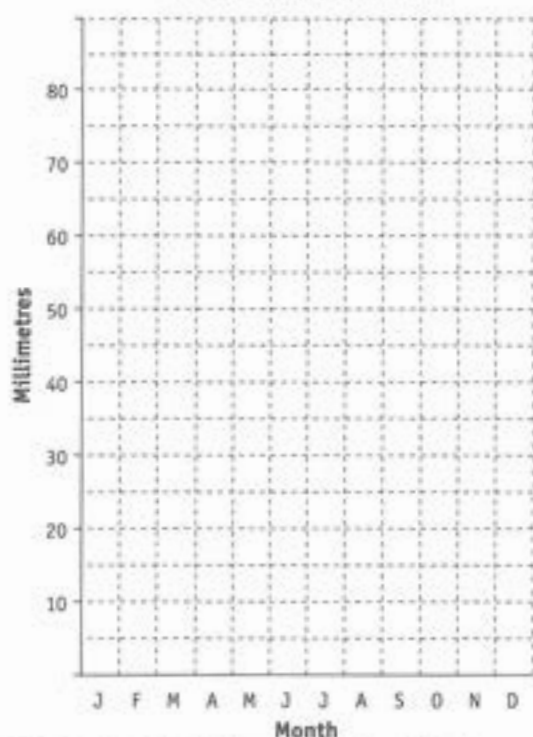
NAME _____ DUE DATE _____

Climate is a pattern of weather. The table below shows average temperatures (in degrees Celsius) and precipitation (millimetres of rain, hail or snow) recorded in the weather station in the town of Gribbsville.

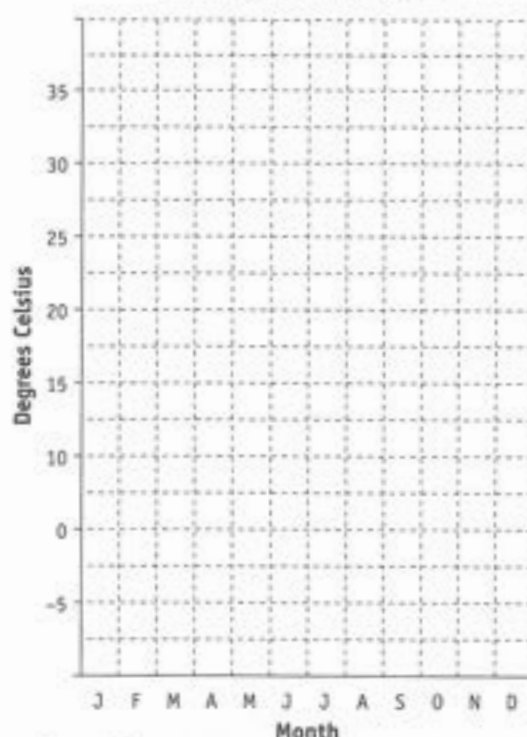
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Temperature (°C)	-4	-3	-1	4	15	18	25	27	21	11	7	4
Precipitation (mm)	28	24	32	41	52	56	74	75	66	39	43	36

- Which month was the hottest? Give the temperature. _____
- Which month was the coldest? Give the temperature. _____
- Which month was the wettest? _____
- Which month was the driest? _____
- What was the total precipitation over the year? _____
- What was the *average* monthly precipitation? _____
- Draw bar charts of precipitation (mm) and temperature (°C) against month in the graphs provided below:

Precipitation in Gribbsville



Temperature in Gribbsville



- Is the town of Gribbsville in the Northern or Southern Hemisphere? Give a reason for your answer.

- What is the difference between 'weather' and climate'? _____

The Earth's Resources

The *biosphere* is the region of the Earth that can support life. This includes the surface of the Earth, the deepest depths of water and out to about 10 km into the atmosphere. You may notice other 'spheres' mentioned in this topic. The *lithosphere* is the solid crust of the Earth and goes to a depth of approximately 80 km. The *hydrosphere* is the part of the Earth where water is present, for example, rivers, lakes, seas and oceans. The *atmosphere* is the envelope of gases surrounding the Earth.

- 1 Give another name for *atmosphere*.

A resource is something that humans find useful for providing food, shelter, clothing and other items needed for living. Air, water, soil, minerals, coal, crude oil and metal deposits are examples of resources. Some resources such as water are *renewable*. This means that they can be recycled in a manner similar to the way water is in the water cycle. Others like coal and oil are non-renewable.

- 2 What does 'non-renewable' mean?

The use of the Earth's resources increased rapidly during the twentieth century.

- 3 Give two reasons why the Earth's resources are being used more quickly than ever before.

Some naturally occurring substances can be used with little processing, for example:

- wood for furniture
- stone for building
- water (containing dissolved solids) for agriculture
- air (a mixture of mainly nitrogen and oxygen) for breathing.

- 4 Give another use for wood.
- 5 Give another use for stone.
- 6 Give another use for water.

Wood has been used to make things and has also been burnt to provide heat energy for thousands of years. Wood can be used to make paper, plywood, particle boards ('craftwood') or as a raw material for the chemical industry. Wood is mainly water and cellulose. Cellulose is a carbohydrate, that is it contains the elements carbon (C), hydrogen (H) and oxygen (O), with hydrogen and oxygen present in the ratio of two hydrogen atoms to every one oxygen atom. Cellulose is a polymer — a large molecule made up of smaller molecules (monomers). The monomers in cellulose are glucose molecules. Cellulose can be digested by some animals but not by humans. Cellulose can be used to make rayon (a synthetic fibre), cellophane (transparent wrapping material), photographic film and explosives.

- 7 What is meant by 'synthetic'?

Wooden articles can last for many years if they are not affected by weather or attacked by insects or fungi.

- 8 What is the *biosphere*?
- 9 Use a labelled diagram to explain the difference between the *lithosphere*, *hydrosphere* and *atmosphere*.
- 10 A definition read, '_____': a substance used to produce heat or power by burning'. What is the missing word?



RESEARCH

- 11 Give another definition for 'resource'
- 12 Find out how wood may be chemically treated to protect it against weather.
- 13 What is involved when wood is 'seasoned'?
- 14 Give another definition for 'polymer'.
- 15 Explain why some animals can digest cellulose, but humans can't.

Building Materials

Stone has been used for thousands of years to make buildings, bridges and works of art.

- 1 Give two reasons why stone is used as a construction material.

Types of stone used for building include sandstone, granite, limestone and marble. The physical characteristics of the type of stone will determine how it is used for building. For example, sandstone is soft and easily cut from quarries; granite is much stronger than sandstone, limestone, and marble, and marble is very popular for making statues and monuments due to its colour and lustre.

- 2 What disadvantage might there be in granite being such a strong stone?

Granite is more resistant to weathering than most other types of stone.

- 3 What does 'resistant to weathering' mean?

Sandstone and limestone are *sedimentary* rocks, that is they are made up of fine grains that have been compressed together over a period of time. Granite is an igneous rock which means it was once molten and has cooled and solidified. Marble is a *metamorphic* rock, that is it has been changed by heat and pressure.

Limestone, chalk and marble are all forms of calcium carbonate (CaCO_3). Limestone is used as a raw material in the manufacture of lime (calcium oxide CaO) and cement. Concrete is made from cement, water (H_2O), sand (mainly silicon dioxide, SiO_2) and small stones.

Concrete is a very common building material. It is strong (and can be made even stronger by adding steel reinforcing), durable, fairly cheap and can be poured into many different shapes.

- 4 What is meant by 'durable'?
- 5 Give one use of marble. Explain what physical property makes it suitable for this use.
- 6 Give two reasons why concrete is such a popular building material.

In areas where wood and suitable stone are not common, brick is used for construction. Nowadays bricks are made in factories from clay, and are baked in an oven, but they have also been made from mud or clay dried in the sun. Bricks can resist weathering, are versatile and have been in use for over 9000 years — the Great Wall of China was built with bricks.

'Wattle and daub' was a type of construction used by early Australian settlers. For this technique wooden poles and twigs are woven together and then plastered with clay.

- 7 Give two reasons why the wattle and daub technique might have been used.

You want to enclose a garden.

- 8 Give two reasons why you might build using wood.
- 9 Give two reasons why you might build using stone.
- 10 Give two reasons why you might build using brick.
- 11 Name one other material that could be used to build a fence or wall around a garden.



RESEARCH

- 12 Find out what types of rock were used to make the pyramids in Egypt.
- 13 Find out when and how cement was invented.
- 14 What type of building material is *adobe*.



Reserves and Resources

Without minerals obtained from the Earth, modern society as we know it would not function. A large proportion of the materials we use are dug out of or pumped from under the ground.

- 1 Name a substance that is pumped out of the ground.

Reserve: the known amount of a substance that is financially worth extracting.

A text book had a passage which stated: 'Total world reserves of copper are thought to be in the region of 360 000 000 tonnes'

- 2 Use your own words to describe what is meant by the passage quoted above.

The same text book stated that: 'At current usage rates, supplies of copper are expected to last approximately 45 years.'

- 3 Identify an assumption made by the person who came up with these figures.
- 4 Calculate the average annual copper usage based on the figures given above. Show your working.

Many text books and Internet sites quote widely varying estimates of reserves of metals. We are not sure of how much of a particular ore is under the ground, nor of its purity.

- 5 What is meant by the 'purity' of an ore?

The amount of reserves may change if we change the rate of use of a material or if new techniques make previously unavailable sources practical.

- 6 Suggest another change that might increase the reserves of an ore.

It is possible that metals may last longer than current estimates because there are probably supplies that have yet to be discovered.

Resource: the total amount of a substance, both known and possible.

Aluminium is a very common metal. Study the table below which shows one estimate of production (millions of tonnes annually, mta) and cost (US\$ per tonne) in the Western world.

Year	1980	1982	1984	1986	1988	1990	1992	1994	1996	1998	2000
Production (mta)	12.0	11.0	11.5	13.5	14.5	13.5	14.0	14.5	15.5	16.0	17.0
Cost (US\$ per tonne)	1750	1650	1300	1200	2600	1140	1200	1600	1700	1500	1200

- 7 Draw a histogram showing how production (y-axis) varies with time (x-axis).
- 8 Draw a histogram showing how cost (y-axis) varies with time (x-axis).

We can make supplies of metals last longer by finding alternative materials.

- 9 Name one other way of making supplies of metals last longer.



RESEARCH

- 10 Find out the current cost of aluminium. Include units of currency and mass.
- 11 Give another definition for 'reserve'.
- 12 Give another definition for 'resource'.

Rocks and Minerals

All of the rocks in the Earth's crust are composed of minerals. A mineral is a chemical compound which occurs naturally in the Earth's crust. There are about 3000 different minerals in the Earth's crust. Minerals can be classified depending on their chemical makeup, crystal structure or physical properties.

Minerals can also be classified into the following four groups.

- **Silicate minerals**, all of which contain the elements silicon (Si) and oxygen (O), and include quartz, feldspar, mica, garnet and talc. About 40% of the known minerals are silicates.
- **Non-metallic minerals**. These might contain a metal but don't look like metals and aren't used as sources of metals. This group includes calcite, sulfur (S), rock salt, gypsum, fluorite and graphite (a form of the element carbon, C).
- **Metal ore minerals** are so named because they are economic sources of metals. The group includes galena (contains lead, Pb), dolomite (contains magnesium, Mg), sphalerite (contains zinc, Zn), bauxite (contains aluminium, Al) and haematite (contains iron, Fe).
- **Gem minerals** which are the precious and semi-precious stones such as opal, jade, garnet, topaz, emerald, ruby, sapphire and diamond (the hardest naturally occurring substance).

Most rocks are made up of more than one mineral. For example, if you have two different samples of granite, they will both contain quartz, feldspar and mica, but the proportions of each mineral will be different. There are so many different minerals that identifying and classifying them is often difficult. Using only one property is not of very much use because, for example, many minerals have the same shape or colour. Several properties are, therefore, used to identify and group minerals. These properties include the colour and the shape of crystals, cleavage (how something splits), lustre (how shiny something is), streak test (what happens when it is rubbed on a hard surface) and hardness.

- 1 What is the main difference between a rock and a mineral?
- 2 Make a list of the elements mentioned in the text above, including their symbols.
- 3 Why is it necessary to classify minerals?
- 4 What are the four main groups into which minerals can be placed?
- 5 Name the minerals used as sources for these metals: iron, zinc and aluminium.
- 6 What minerals are found in granite?
- 7 What is the hardest mineral?
- 8 Name three properties that can be used to help identify minerals.



RESEARCH

- 9 Why are coal, pearls and amber not classified by some mineralogists as true minerals?
- 10 Explain how mineralogists use Moh's scale of hardness.
- 11 Find out two uses of graphite.
- 12 Give an alternative definition for 'mineral' to the one used earlier.
- 13 What gemstones are found at Lightning Ridge?

Ores and Metals

Ore: a mineral containing enough of a metal to make it worthwhile mining.

Ores consist of metals chemically combined with other elements — usually oxygen (forming an oxide) or sulfur (making a sulfide). Some metals can be found *native*, that is, uncombined. Gold (Au), silver (Ag) and copper (Cu) are examples of native metals. Jewellery, ornaments and coins have been made from silver and gold for a very long time. Both are attractive metals, resist corrosion, are relatively scarce and are therefore, valuable.

The amount of metal ores in the Earth's crust is limited. We cannot continue digging them up forever. Sooner or later there will not be enough to be worth mining. The amount of metal ore we *know* is worth getting out is called the *reserve* of that ore. The table below shows one estimate of approximately how long metal ores will last if we continue to use them at the present rate.

Metal	Years
Aluminium	250
Copper	50
Chromium	110
Gold	30
Iron	200
Zinc	30

The metals are likely to last longer than these figures indicate. There are probably supplies which we have not yet discovered. Recycling is one way of making metal supplies last longer. This means that metals are melted down and used again. Over 90% of gold we use is recycled, though this only happens to 50% of iron (Fe). We can do more to recycle metals now, and in the future we will need to do so. It has been estimated that the average family of four will throw away about 25 kg of aluminium (Al) and 70 kg of iron per year.

Some metals are more abundant than others. The prices of metals depend on how plentiful they are, how much a demand there is for them and, also, how hard it is to extract them from their ores. The more reactive metals are more difficult (and expensive) to obtain from their ores. Iron and aluminium are two important elements found as oxides. Haematite is a type of iron oxide and bauxite is impure aluminium oxide. The pure metals are obtained by removing the oxygen. Iron is obtained from its ore by using a blast furnace, and electrolysis (splitting the ore using electricity) is used to extract aluminium.

- 1 What elements are present in haematite?
- 2 What elements are found in bauxite?
- 3 What is meant by the *reserve* of an ore?
- 4 Assuming that the population of Australia is 17 million, work out how many tonnes of iron and aluminium are thrown away annually.
- 5 When are the supplies of copper extracted from the ground likely to run out?
- 6 Explain why the supplies of copper might last longer than estimated.
- 7 Explain why the supplies of copper might run out sooner than estimated.
- 8 Give a reason why much more gold than iron is recycled.
- 9 Name three factors involved in the cost of a metal.
- 10 List the names and chemical symbols of the metals mentioned on this sheet.

Iron and Steel

With the exception of certain meteorites, iron (Fe) does not occur *native* (uncombined with other elements) in the Earth's crust. There are many compounds of iron, such as iron sulfide (iron pyrites, FeS_2), haematite (Fe_2O_3) and magnetite (Fe_3O_4). Iron is the second most abundant metal in the Earth's crust. Because it can be relatively easily extracted from its ores and because it has useful physical properties, it has become a metal of considerable importance.

Iron is produced in a *blast furnace*. This is a tapered, cylindrical tower approximately 30 to 40 metres high. It is constructed of steel and is lined with special heat resistant (refractory) bricks.

A blast furnace works continuously and can produce more than 1000 tonnes of iron every 24 hours. The raw materials are iron ore (for example, haematite), coke (carbon, C), limestone (calcium carbonate, CaCO_3) and air. Coke is obtained from coal.

The molten iron formed flows to the bottom of the blast furnace. Impurities (mainly calcium silicate) form as *slag*, floating on top of the molten iron. Slag and iron are tapped off, and allowed to cool and solidify. The iron produced is still relatively impure and is known as *cast iron* or *pig iron*. It contains about 4% carbon. It is very hard but brittle. This type of iron is only used for objects which are not likely to be bent when used.

Pure iron is not a good material for building things. It corrodes (rusts) very easily and is soft. Most iron produced in blast furnaces is turned into steel, an alloy (mixture) of iron. There are thousands of types of steel and their properties will depend on the other elements added. The impurities present are removed from molten cast iron in a special furnace and other elements are added in careful proportions. Mild steel is commonly used for car bodies. It is cheap, easily shaped and relatively strong.

- 1 What does 'native' mean?
- 2 List the names and formulae of all of the iron compounds mentioned on the sheet.
- 3 Give the formula for calcium carbonate.
- 4 Give the formula for carbon dioxide.
- 5 Give the formula for carbon monoxide.
- 6 What does 'refractory' mean?
- 7 Iron is a 'non-renewable resource'. What does this mean?
- 8 What are the raw materials used in a blast furnace producing iron?
- 9 What is 'slag'?
- 10 Would you use cast iron to make a bridge carrying heavy traffic? Give a reason for your answer.
- 11 What happens when a metal *corrodes*?
- 12 Why might mild steel be used in car bodies?
- 13 Give three considerations to be taken into account when siting a blast furnace.



RESEARCH

- 14 You may have heard of 'The Iron Age'. When did this occur and what did it involve?

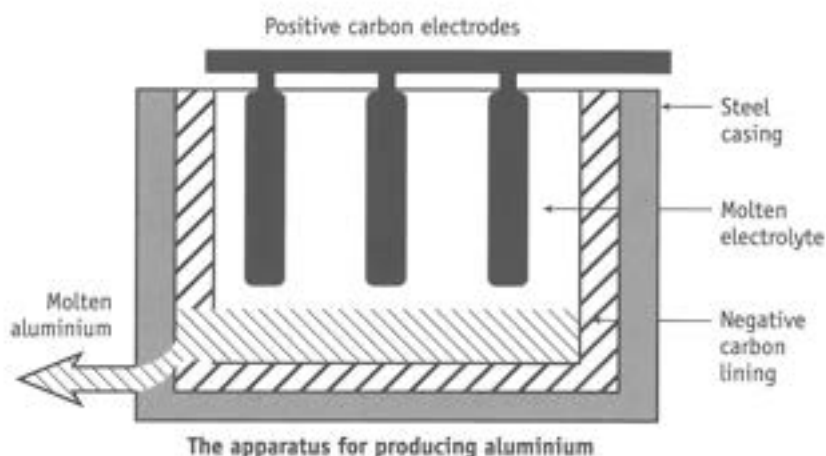
Aluminium

Using an electric current to split up substances is called *electrolysis*. This process is widely used by industry, for example, to produce hydrogen (H), chlorine (Cl) and sodium (Na). Electrolysis is also used in the refining of gold (Au), copper (Cu) and silver (Ag). Aluminium (Al) is the most abundant metal in the Earth's crust (about 8% aluminium compounds) but, until about a hundred years ago, when electrolysis was first used industrially, aluminium was rarer than gold. The principal ore providing aluminium is bauxite. Bauxite is an impure form of aluminium oxide (Al_2O_3). It must be purified before it can be used.

Aluminium is a very reactive metal and it takes a lot of energy to separate it from oxygen. Aluminium oxide has too high a melting point to be electrolysed (split by electricity) easily in its pure form. A mineral called cryolite is melted and the alumina is dissolved in this. This lowers the melting point (from approximately 2000°C to around 900°C). Molten aluminium collects at the bottom of the cell and is drained off, cooled and solidified. Oxygen (O) gas is given off at the positive carbon (C) electrodes.

This is a continuous process, and raw materials are added at regular intervals. The high current used (up to 125 000 amps) keeps the electrolyte (substance being electrolysed) molten. Aluminium can only be produced economically when large amounts of cheap electrical power are available.

Aluminium is used where weight is important. When alloyed with other metals (such as magnesium, Mg) it is strong. Train and aeroplane bodies are made from aluminium alloys. Aluminium is also a good conductor of electricity and can be used as an alternative for (the more expensive) copper. Aluminium resists corrosion (rust) — it has a tough coating of unreactive aluminium oxide on its surface.



- 1 List the names and symbols of all the elements mentioned on this sheet.
- 2 Give the name of the main constituent of bauxite.
- 3 Define 'electrolysis'.
- 4 Copy the diagram of the apparatus used to produce aluminium from aluminium oxide.
- 5 What are the two products of the electrolysis of the purified ore?
- 6 Explain why aluminium can be collected from the bottom of the electrolytic cell as shown in the diagram.
- 7 Give one drawback of using electrolysis to extract aluminium from its ore.
- 8 Why might aluminium be used to make aircraft bodies?
- 9 Copper is a better electrical conductor than aluminium. However, aluminium is sometimes used instead of copper for electrical wiring. Why might this be so?
- 10 Give one example of when aluminium might be used instead of copper as an electrical conductor.
- 11 What percentage of the Earth's crust is aluminium?
- 12 What properties might make aluminium foil useful in cooking?



RESEARCH

- 13 Many metals are extracted from their ores by heating with carbon. Why is this method unable to be used with aluminium?
- 14 Name one place in Australia where bauxite is mined.

Glass

Glass has been used for over 4000 years to make containers and ornaments. More recently it has been used for windows. Glass is a poor electrical and thermal conductor. Glass is neither a 'proper' solid nor a liquid but is a 'super cooled liquid'. Over hundreds of years, glass can very slowly flow and change shape. The most commonly used glass (used in windows, jars, drinking vessels and so on) is *soda glass*. The raw materials used in the manufacture of this type of glass are sand (silica, silicon dioxide), sodium carbonate ('soda') and calcium carbonate (limestone).

'Crystal' glass contains lead oxide, is heavy and refracts (bends) light well. It is used in making 'costume' jewellery, lenses and prisms. Glasses containing lead are used to protect people working with radioactivity.

Laboratory glassware is made of borosilicate glass (Pyrex™) and contains borax. This is also used to make cooking containers. It is strong, heat and chemical resistant.

Vehicle windshields are made of laminated glass. This consists of two sheets of thin plate glass with a layer of plastic in between. The glass used to make lenses in spectacles, cameras, microscopes, and telescopes needs to be very pure and free of distortion. Although heavier than plastic, and more likely to shatter when dropped, glass does not scratch as easily as plastic.



Fibre optics is the branch of optics (the study of light) dealing with the transmission of light through fibres or thin rods of glass. Light can travel through the fibre over long distances with very low loss, even if the fibre is curved. Optical fibres are used in medical instruments for viewing inside the human body and for laser surgery. One growing application of optical fibres is in communication, where cables can carry much more information than the equivalent copper cable.

Glass fibres can be woven in the same way cotton and wool can be woven into cloth. This type of material can be used for fire retardants or can be mixed with polymers (plastics) to make a composite material called glass reinforced plastic (GRP), sometimes called *fibreglass*. This is a light and strong material that combines the best properties of both materials.

- 1 What are the main ingredients for soda glass?

Clear plastics can be used as a substitute for glass.

- 2 Give two ways in which clear plastics are better than glass.
- 3 Give two ways in which clear plastics are not as good as glass.
- 4 Glass is said to *refract light*. What does this mean?
- 5 Why is borosilicate glass used to make the beakers and flasks used in chemistry laboratories?
- 6 For approximately how many years has glass been used by people?
- 7 Boats and surfboards are often made out of GRP. Give two reasons why GRP is a suitable material for this purpose.
- 8 Why are copper telephone wires being replaced by fibre optic cables?
- 9 Give one use for glass that has not previously been mentioned on this sheet.



RESEARCH

- 10 What is meant by the words in **bold** in the following sentence? 'Usually **transparent**, glass can also be **translucent** or **opaque**.'
- 11 What does 'vitreous' mean?



Paper

Human beings have kept records since at least the last Ice Age, about 20 000 years ago. Many systems of characters and symbols have been used. The Sumerians (who lived in a region similar to modern day Iraq) pressed wedge-shaped marks into wet clay about 5000 years ago, using a language known as *cuneiform*. Other civilisations have used marks on parchment (goat or sheep skin) or vellum (calfskin). The ancient Egyptians wrote on papyrus, a plant-based material. Paper was invented by the Chinese nearly 2000 years ago. Paper can be produced from almost any plant material fibre containing *cellulose*. Wood pulp is the most common (95% of total production) but bamboo, cotton (and similar fibres) and straw can also be used.

Here is a simplified sequence for making paper:

- wood is made into small wood chips
- chips are turned to pulp by being ground and/or by being treated with chemicals
- the pulp is bleached
- substances are added to the pulp to give paper colour, texture and lustre
- water is added
- the pulp is passed through rollers, where the water is removed by squeezing and heating
- the finished paper is wound on reels.

To make paper, a large amount of raw materials is needed. It also requires a large amount of energy and generates a large amount of effluent (waste). Many people are concerned with the chemicals (often based on chlorine) used to bleach paper. *Environmentally friendly* brands of paper have been produced which are either not bleached or are bleached using oxygen.

Recycling paper can help cut down on the amount of wood needed for paper. However, recycled paper cannot be used by itself to make high quality paper. It needs to be either used to make packaging materials, or to be mixed with new wood pulp. Before printed paper can be recycled it has to be 'de-inked'. Australian industry recycles a large quantity of the paper it uses.

- 1 How long have humans been using paper?
- 2 Name two substances, apart from paper, that people have written on.
- 3 What is 'effluent'?
- 4 What is the major source of cellulose used in making paper?
- 5 Draw a flow chart showing how paper is made from trees.
- 6 Apart from sheets of paper used for writing, name three other products made from paper.
- 7 Why might some environmentally conscious consumers object to coloured toilet paper?
- 8 Give two reasons why the recycling of paper can be seen as desirable.
- 9 A roll of newsprint can be approximately 170 cm wide by 8000 m long. What is the area of this paper in square metres (m^2)? Show working. (Hint: check units)



RESEARCH

- 10 *Dioxins* have been associated with chlorine-based bleaches. Find out what problems have been associated with dioxins.
- 11 Find out the processes involved in recycling paper.

Mining and Minerals: Assignment

NAME _____ DUE DATE _____

- 1 On this map of Australia, draw in the States and Territories.
- 2 On the map, show where these important mining sites are:
Broken Hill (NSW)
Mt Isa (QLD)
Kalgoorlie (WA)
King Island (TAS)
Tennant Creek (NT).

You may use a key.



- 3 Give the name of one substance mined in each of these locations:

Broken Hill _____ Mt Isa _____

Kalgoorlie _____ King Island _____

Tennant Creek _____

Gold is mined in Australia.

- 4 What is the chemical (shorthand) symbol for gold? _____
- 5 Name two uses for gold. _____
- 6 Give the name of one place where gold is mined in Australia. _____
- 7 Show your answer to Question 6 on the map.
- 8 For what is Coober Pedy famous? _____
- 9 In which State/Territory is Coober Pedy? _____
- 10 Give a definition for 'ore' _____

- 11 What metal is obtained from bauxite? _____
- 12 What metal is obtained from haematite? _____



Balance in Nature

Ecosystem: a space containing organisms interacting with each other and their surroundings.

If an ecosystem has existed successfully for a long time, then the populations (members of a particular species) are usually stable. This occurs when all of the organisms in an ecosystem are reasonably compatible with each other and their environment. If numbers do change, this often happens at regular intervals in a way that can be forecast, for example, in line with changes of seasons.

An ecosystem can be upset by the arrival of a new species or the loss of an existing one.

1 Apart from living creatures, what factors can change an ecosystem?

Cattle were first brought to Australia by sailing ships late in the eighteenth century. Due to the long voyages many **organisms** were either left behind or died on the way. These included both **beneficial** and **pathogenic** creatures. The dung beetle was one example. Dung beetles lay eggs in cattle **dung**. When the **larvae** hatch out they eat the dung. This makes sure that the compounds present in the dung are returned to the soil. It also keeps down the number of flies (which also **breed** in the dung) and increases the area available for **pasture**. There are dung beetles **native** to Australia, but these tend to eat native wildlife dung and do not eat the dung of introduced cattle.

2 Give meanings for the words shown in **bold** in the passage above. Choose from: *disease causing* *helpful* *excrement* *living creature* *indigenous* *grass* *young insects* *produce offspring*

If cattle are introduced without dung beetles then they are likely to have an adverse (hostile) effect on their new environment. Because flies tend to eat dung, their numbers will increase — with the obvious consequences. Some environmental organisations monitor the number of dung beetles. Dung beetles are killed by some cattle drenches (medicine). It has been suggested that different species of dung beetles should be introduced from overseas.

3 Give two potential effects of introducing different species of dung beetles to Australia.

For the following questions assume that:

- there are 40 million cattle in Australia
 - each cow deposits 10 pats of dung per day
 - the average area of a cow pat is 500 square centimetres (cm^2)
 - there are 10 000 cm^2 in a square metre (m^2)
 - there are 10 000 m^2 in a hectare.
- 4 Calculate the area of pasture one cow covers with cow pats per day. Express your answer in cm^2 and in m^2 .
 - 5 Calculate the total area of pasture covered in Australia per day. Express your answer in m^2 and in hectares.
 - 6 Assuming that none of the dung was broken down, what would be the total area of pasture covered in a year? Give your answer in hectares.
 - 7 How do dung beetles ensure that compounds are returned to the soil?
 - 8 Apart from dung beetles, what might break down cow pats?
 - 9 Are cattle a species native to Australia? Explain your answer.
 - 10 Define 'population'.
 - 11 Define 'ecosystem'.



RESEARCH

- 12 Give a more detailed definition of 'drench'.

Components of Ecosystems

The *environment* is the conditions in which a plant or animal lives. An environment consists of living (biotic) and non-living (abiotic) components. Abiotic components include things like air temperature, soil pH, and so on.

- 1 Name two other possible abiotic components in an environment.

Ecology is the study of living things and their environment. The place where a plant or animal lives is its *habitat*. There are many types of habitat in Australia, such as ocean and rainforests.

- 2 Name and describe two other types of habitat in Australia.

Population means all the members of a particular species (related organisms) found in an area. An *ecosystem* is a group of living things together with their habitat or environment. You rarely find a single type of organism living on its own. Populations of different species can be found sharing a habitat. There is usually interaction with other species. When there is a collection of different species that exist with others, this is a *community*. Koalas, for example, breed and live in the trees.

- 3 What do eucalypts provide for koalas?

The trees, in turn, need water, sunshine and nutrients from the soil. The soil is made up of the remains of plants and animals that have rotted down. There are three categories of populations: *producers*, *consumers* and *decomposers*.

Green plants are producers. They are the first link in all food chains. They contain chlorophyll and so can *photosynthesise*. This process produces a food called glucose (a type of sugar) and also oxygen. It uses the energy of the sun to drive the reaction. This solar energy is converted to stored chemical energy in plants.

- 4 What would happen to life on Earth if all the green plants suddenly died?

Animals are consumers. If they eat plants, they are first-order consumers or *herbivores*. The chemical energy in the green plants is transferred to the animal when it eats the plant.

Consider cows (first-order consumers) grazing in a grass paddock. Typically 1000 megajoules (MJ) of energy from the sun falls on one square metre (1 m^2) of ground per year. It has been estimated that 500 MJ is used up evaporating water, 320 MJ goes into the ground, 177 MJ is reflected back, 21 MJ is used for new plant growth, 2 MJ is used for respiration and 3 MJ goes to the cow.

- 5 Construct a table showing the breakdown of how the energy falling on a square metre of grass annually is used. Work out the percentage of total energy which is received by a cow.
- 6 Define 'ecology'.
- 7 What is a 'community'?

What do the following definitions describe?

- 8 An organism which produces its own organic compounds from simple raw materials (such as carbon dioxide and inorganic nitrogen).
- 9 A biological classification consisting of related organisms potentially capable of interbreeding.
- 10 An organism that needs complex organic compounds for food, which it obtains by eating other living things.



RESEARCH

- 11 Find out what a *biome* is and give three examples.
- 12 What is the typical energy intake of an adult cow per day?

Components of Ecosystems

Animals which eat other animals that are herbivores are second-order consumers or *carnivores*. They in turn can be eaten by (usually larger) third-order consumers. Animals which eat both plants and animals are called *omnivores*. We can show the relationship between producers and consumers using a *food-chain*:

- 13 What does '→' mean in the food chain? leaves → caterpillar → bird → cat

If one part of a food chain or food web is upset then the whole system will be affected.

Populations can be changed by disease, detrimental weather conditions and change in numbers of consumers above or producers below in the food chain.

Here is a simplified marine food chain: plankton → krill → whale

Plankton are microscopic organisms, including bacteria, fungi, algae and the eggs and larvae of many different animals. Krill are small (1 cm to 7 cm long) shrimp-like crustaceans which live in dense swarms. They are the chief food of many marine animals. They have even been fished for human consumption because they are high in protein. If too many krill are used by people then:

- too much plankton may cause problems in the sea if no other animals eat them
- those whales which eat only krill will die.

- 14 What is meant by 'marine'?
 15 What are the producers in the previous food chain?
 16 What might hunt or eat whales?

Study this food chain: green plant → caterpillar → dragonfly → snake → hawk

If the numbers of dragonflies were to lessen, decide what would happen to the other producers and consumers.

- 17 What would happen to the number of caterpillars?
 18 What would happen to the number of snakes?
 19 What would happen to the number of green plants?
 20 Is a food chain a complete description of 'who eats what'? Give an explanation.

In a food chain, chemical energy moves from one organism to another. The last part of the chain involves decomposers. These are things such as bacteria and fungi which rot down (decompose) dead animals and plants and return their chemicals and other components to the system.

- 21 What would happen to an ecosystem if it didn't contain any decomposers?
 22 Name three things that can affect the number of organisms present in a population.

When many food chains are joined together to show the relationship between producers and consumers this is called a *food web*.

- 23 Explain the difference between 'consumers' and 'carnivores'.



RESEARCH

- 24 A text book described krill as 'an important component of pelagic fauna'. Explain what is meant by 'pelagic fauna'.
 25 Find out two facts about krill which are not mentioned on this sheet.
 26 Find out two facts about plankton which are not mentioned on this sheet.
 27 Do all whales eat krill or is it just a few species?



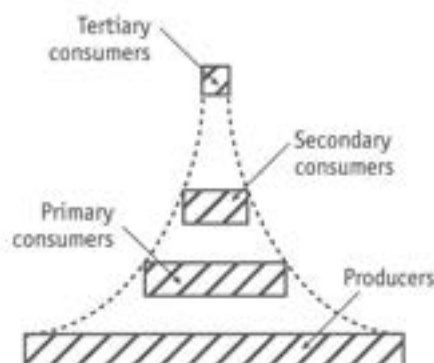
Food Pyramids and Energy

For a population, or any life, to exist there must be adequate food supplies and shelter. The size of the population will tend to grow to whatever the habitat will support. How quickly the organism reproduces is also important. We use *food chains* and *food webs* to show how organisms are linked in ecosystems. At each step in a chain, energy is 'lost' — only part of the energy is available to the next organism. Less than half the energy taken in by a cow by eating grass will be available when the cow itself is eaten.

- 1 Where did this 'lost' energy go?

Trophic (nutritional) levels describe the numbers of organisms in food chains using energy levels. Organisms in upper levels feed on the organisms in the level below.

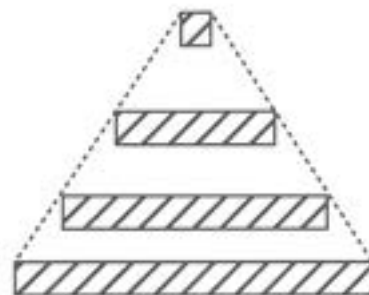
- 2 Copy the diagram on the right into your book.
- 3 Show on the diagram where the following should be placed:
vegetation carnivore
herbivore secondary carnivore.
- 4 This 'pyramid of numbers' shows that the number of organisms decreases as we go up the pyramid. Explain why this happens.



Biomass: the abbreviation for biological mass, the amount of living material in a particular system.

Another way of showing how things are positioned in a food chain is to use a pyramid of biomass.

- 5 Copy the diagram opposite into your book.
- 6 Label the diagram with:
producer first consumer
second consumer third consumer
- 7 Assuming that only 10% of biomass is transferred at each stage, what percentage of energy originally present in the producer would reach the third (tertiary) consumer? Show your working.
- 8 There are fewer numbers of animals in higher levels, but this doesn't match the pattern of biomass in each level. What happens to the size of animals (in general) as we go up levels?
- 9 Why do you think that there are very rarely more than four or five links in a food chain?
- 10 Name three things that will affect the size of a population.



What is described by each of the following definitions:

- 11 A level of a food web containing organisms arranged in order from primary producer to final consumers.
- 12 A community of organisms and their environment functioning as an ecological unit in nature.



RESEARCH

- 13 Briefly explain what is meant by the term 'biomagnification'.

Interactions Between Organisms I

We are familiar with how groups containing the same species of animal can live together and cooperate, especially in social groups. One interesting example is provided by the Southern African meerkats — small mammals which help each other in several ways. Adults in a troop share 'babysitting' duties when mothers go hunting, even if they are not related to the young. They are notable for having 'lookouts', who stand on their back legs, using their tails for support, and give the alarm if predators are seen.

- 1 Why is sharing duties helpful to the group?
- 2 Name two other advantages that living in a group can give to its members.

However, it is not so obvious that different species can also live together in associations which can benefit them. The word *symbiosis* comes from the Greek, meaning 'living together'.



Symbiosis: the interdependence of two organisms not of the same species.

If symbiosis results in both of the organisms gaining from the relationship, this is called *mutualism*.

Algae and fungi are classified as lichen. Lichen is found in damp places. The algae (being capable of photosynthesis) can make food for both themselves and the fungi. The fungi store water and protect the algae.

- 3 What process is likely to be used by algae to make food?

There are several types of relationships occurring between organisms. Another example is *commensalism*. Commensalism means 'eating from the same table', and, as the name suggests, it involves sharing.

Commensalism: a relation between two types of organisms in which one obtains benefits from the other without harming it.

Consider mice in a house. They eat any food that is left around and take shelter in the dwelling.

- 4 Do humans gain from this relationship? Explain your answer.

Parasitism involves one organism (the parasite) living in or on another (the host) and taking food from it. This usually harms the host.

Parasitism: a close association between organisms in which one organism obtains benefits from another which it often damages.

Lice and fleas are parasites. They exist on humans by sucking blood.

- 5 Give another reason why the presence of lice and fleas is considered unhealthy.
- 6 Name another insect that can suck blood from humans.



RESEARCH

Find out two examples for each (not mentioned on the sheet) of:

- 7
 - mutualism
 - commensalism
 - parasitism.

Interaction Between Organisms II

Before starting this sheet, you will need to make sure you know what is meant by each of the following:

- symbiosis
- mutualism
- commensalism
- parasitism.

1 Define 'symbiosis'.

Read each of the following scenarios and decide into which category (mutualism, commensalism or parasitism) they fall. Explain each answer.

- 2 Wildebeest are large antelopes found in southern Africa. They are grazers. Like many animals they are troubled by insects. These large horned mammals allow oxpeckers (birds) to sit on them and eat annoying insects.
- 3 Remora have flat disks behind their heads which act as suction devices. They attach themselves to sharks and feed on food scraps left by their hosts. They can also swim by themselves.
- 4 Tapeworms, found in the intestines (bowels) of some vertebrate animals, can be from 10 mm to 9 m in length. They absorb partially digested food through their exterior. They reduce the fitness of the host.
- 5 Plants that have flowers produce nectar (a sugary liquid) to attract insects to feed. When insects visit a flower, they are dusted with pollen and carry this away to other flowers that they pollinate.
- 6 Plants need nitrogen to be 'fixed', that is supplied in a soluble form. Rhizobium bacteria can form lumps on the roots of plants such as alfalfa, beans, clover and peas. These plants supply food (carbohydrates) to the bacteria and receive useful nitrogen compounds in return.
- 7 Some ferns are *epiphytes*, which mean they do not have roots in soil but, instead, attach themselves to trees. They take no sustenance from the tree but live on moisture and particles which they absorb from the air through specialised leaves and roots.
- 8 The carob (or locust bean) has edible fruit in pods containing seeds. The seeds are not viable (will not grow) until they have been eaten and passed through the digestive system of an animal.
- 9 Birds carry mistletoe seeds from tree to tree. The mistletoe never forms roots of its own. The mistletoe plant develops growths which penetrates the tree and take nourishment from the host plant.



RESEARCH

- 10 Find out two additional facts about the carob tree (*Ceratonia siliqua*).
- 11 What is a gnu?
- 12 Find out three facts about lampreys.

A Population Explosion



An example of *population explosion* is the rapid growth in the number of people on Earth due to people living longer and more babies being born.

Study the table below that shows estimates of the Earth's human population.

Year	Population (millions)
8000 BC	5
4000 BC	75
2000 BC	150
AD 1 (Birth of Christ)	250
AD 1650	500
AD 1830	1000
AD 1930	2000
AD 1950	2500
AD 1960	3000
AD 1970	3600
AD 1975	4000
AD 1990	5300

The first person to go on record with concerns about the growth in human population was Thomas Malthus. *Demography* is the study of how human populations change with time, and this branch of science came about as a result of Malthus' researches on the increase in human numbers. He asserted that the population would naturally increase more quickly than the supply of food that could be grown to keep the population alive. In other words, the demand for food would outstrip supply. When this happened starvation and disease would rapidly reduce the numbers to a more manageable level, until they rose again.

- 1 Draw a graph showing how the human population has varied over the last 10 000 years. Make sure your time scale extends to AD 2100.
- 2 On your graph indicate which is the 'dependent' variable and which is the 'independent' variable.
- 3 Use your graph to estimate the size of the population in 1000 BC.
- 4 Estimate the size of the population in AD 1000.
- 5 Estimate the size of the population in AD 1900
- 6 Show how you obtained your results for Questions 3, 4 and 5.
- 7 Extrapolate your graph to forecast the population in AD 2100.
- 8 Name one assumption you made in your answer for Question 7

Study the size of the population in the following years. Work out how long it took for each of the following populations to double in numbers.

- 9 How long did it take the population in 2000 BC to double?
- 10 How long did it take the population in AD 1 to double?
- 11 How long did it take the population in AD 1900 to double?
- 12 Where on your graph is population increasing the quickest?
- 13 Will it be possible for the rate of increase to be maintained? Give a reason for your answer.
- 14 Give two reasons why the size of human populations might not behave in the same way as those of plant or animal populations.

Pest Control

Pests cause major damage to crops all over the world. Pests include insects, plants, animals and even micro-organisms. It is thought that pests destroy around one third of all produce, costing billions of dollars when many areas are short of food. Many introduced species are pests, for example rabbits, foxes and 'Patterson's Curse'.

- 1 Name one bird which eats a food crop.
- 2 Name one insect which eats a food crop.

Farmers try to reduce the pest population by chemical, physical, and biological means. *Pesticides* are chemicals that include *herbicides* (to kill weeds) and *insecticides*.

- 3 What does an *insecticide* do?

Not all crops are treated in the same way. Cotton, corn and soy beans are heavily dependent on pesticides when grown commercially. Pesticides are efficient, with only a few kilograms per hectare needed.

- 4 What is meant by: 'Farmers may get a 400% return on money spent on insecticides.'

However, pesticides, although useful, can also cause serious problems. Insecticides can kill non-target species, such as harmless insects. These insects may pollinate other plants or be the food for birds.

- 5 Describe two more disadvantages of using pesticides.

Weeds can be pulled from the soil by hand or physical barriers (such as mulch) can be used.

- 6 Explain how mulch stops the growth of weeds.

Many biological controls have been tried as more selective alternatives to pesticides. Introduced pests have been controlled by also introducing their predators and parasites.

- 7 What is meant by saying that biological controls can be 'more selective' than pesticides?

This approach has been successful in some areas, for example, bringing in a moth (*Cactoblastus cactorum*) from South America to feed on the out-of-control prickly pear cactus (*Opuntia strictia*). Diseases can also be used, but the pest can eventually develop a resistance to the pathogen.

Read the following abbreviated extract from a newspaper article and answer the questions that follow.

Rabbits make a comeback

Local authorities have expressed concern about recent increases in the numbers of rabbits. Seemingly the calicivirus which was introduced two years ago, and had a huge effect on the

rabbit population, has lost its initial impact.

Farmers have been reminded of their obligation to control the numbers of rabbits on their properties. A spokesman said, 'Calicivirus still has its uses, but we can't just rely on one means of control. People may have to go back

to more traditional ways of keeping rabbit numbers down until new strains of calicivirus are developed or occur naturally'.



- 8 Explain why calicivirus may no longer be killing rabbits.
- 9 Name two problems that are caused by rabbits.
- 10 State two ways in which farmers can control rabbits without relying on calicivirus.



RESEARCH

- 11 Find out why the cane toad (*Bufo marinus*) was introduced to Australia.
- 12 Find out a biological control method used to keep insect numbers down.
- 13 Find out what DDT is and why it is no longer used.

Ecology: Cloze

Copy out the sentences and fill the numbered gaps using words from this list:

biomass	carnivores	community
consumers	decay	decomposers
eaten	ecology	ecosystem
environment	food	green
habitat	herbivores	incorporated
mutualism	organisms	parasitism
photosynthesis	plants	population
producers	relationship	solar
sun	symbiosis	system
temperature	warm	web

Each word is used once.

All living (1) need energy. The (2) provides energy for green plants to grow. Green plants use this (3) energy and change water and carbon dioxide into food and oxygen by a process called (4). These plants are called (5) because they make their own food. Animals can eat (6) to gain energy and, in turn, can be (7) by other animals. Of all the sun's energy that falls on Earth, only 1% is (8) into plants. Then, as you go up the (9) chain, about 90% of the available energy is used because animals need to run, fly, hop and keep their bodies (10).

Organisms that don't produce their own food, but that eat other living things are called (11). Animals that eat plants are called (12), those that eat other animals are (13), and omnivores eat both plants and animals. Food chains usually start with a (14) plant. Some organisms (such as bacteria and fungi) cause (15) and help recycle the energy locked up in living things. These are called (16). When many food chains are joined together to show the relationship between producers and consumers this is called a food (17). If one part of a food chain or food web is upset, then the whole (18) will be affected.

The (19) is the conditions in which a plant or animal lives. (20) is the study of living things and their environment. The place where a plant or animal lives is its (21). A group of organisms of the same species is called a (22). A collection of organisms of different species co-existing is called a (23). An (24) is a group of living things interacting together with their habitat or environment. Living things and their environment affect each other. Important environmental factors include (25), light, water and the type of soil.

(26) is the interdependence of two organisms not of the same species. If symbiosis results in both of the organisms gaining from the relationship, this is called (27). Commensalism is a (28) between two types of organisms in which one obtains benefits from the other without harming it. (29) is a close association between organisms when one organism obtains benefits from another which it often damages.

Energy I

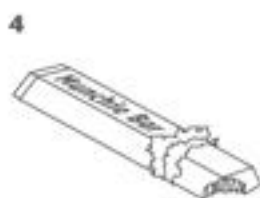
- 1 We often talk about energy. Before reading any further, write briefly (no more than two or three lines) what you think 'energy' means.

Now study this definition and compare it with your version.

Energy: the ability to do work or to change things.

Energy exists in many different forms, for example, light, sound, heat, electrical, movement (kinetic) and potential (stored) energy. There are various types of stored energy, such as chemical, positional and elastic. Foods and fuels contain chemical energy. This energy is released when foods are digested or fuels are burned. Objects high from the ground have positional energy (another form of potential energy) — this is changed to movement (kinetic) energy when they fall. Stretched springs (or elastic bands) have elastic energy, this can be changed to movement — clockwork toys and catapults are examples.

Name the type of stored energy shown in the following six situations. (Not to scale.)



As we have seen, energy can be changed from one form to another, for instance a light globe changes electrical energy to light (and heat) energy, a drum changes movement energy to sound energy, a barbeque changes chemical (potential) energy to heat (and light), and so on.

- 7 Can you see energy?
- 8 Is it possible to weigh energy?
- 9 Does energy take up space?

We never get all of the stored energy out in the form that we want it, some is always changed to other forms of energy. In a car engine, only some of the stored energy in petrol goes to movement energy. The rest is changed to other types of energy, such as heat and sound.

- 10 Consider a cyclist. His or her energy is supplied by food. As he or she is cycling along, is all of the energy being turned into movement of the bicycle? Explain your answer.

Cells (batteries) are sources of electrical energy. They have stored (chemical) energy that can be changed to electrical energy.

- 11 State an advantage of having an electrical appliance (machine) that works on batteries rather than mains electricity.
- 12 State a disadvantage of having an electrical appliance that works on batteries rather than mains electricity.
- 13 Give three examples that have not been mentioned of objects with kinetic energy.



RESEARCH

- 14 Albert Einstein came up with a famous equation, $E = mc^2$. Find out what this equation means. How has this equation been applied in making weapons?
- 15 How do batteries (cells) act as a source of electrical energy?
- 16 Explain how a conventional filament-type globe changes electricity to light.

Energy II: Assignment

NAME _____ DUE DATE _____

ENERGY CONVERSIONS

Energy makes things happen. It can be transformed (changed) from one type to another. One example is a light globe. When electricity passes through the metal filament, the wire heats up, glows and gives off light (and heat). Look at the following five diagrams and say what energy changes are occurring (in other words what type of energy is first supplied and to what type of energy it is being changed).



- 1 _____
- 2 _____
- 3 _____
- 4 _____
- 5 _____

Energy is needed to change things. Energy is involved if something moves, gets warmer or colder, changes shape, gives off light and so on.

- 6 Someone says that energy can be 'used up'. Do you agree with this? Explain your answer.

7 Name a machine that changes chemical (stored) energy to movement. _____

One joule (J) is the unit of energy equal to the work done by a force of one newton (N) acting through a distance of one metre (m). It is named after James Prescott Joule, an English physicist.

Energy is measured in units called joules (J). The joule is a small unit so we often use kilojoules (kJ).

- 8 How many joules are in a kilojoule? _____

Packaged food often has details of the energy content (in joules) shown on the label.

- 9 Why might people want to know the energy content of foods?

- 10 Name three fuels. _____

Fuel: a material that is used to produce heat or power by combustion (burning).



RESEARCH

- 11 What does the following sentence define? 'The force needed to give an acceleration of one metre per second per second to a mass of one kilogram'

Energy from Biomass

With fossil fuels running out, the world needs *renewable* energy sources — energy sources that can be made again as fast as they are used up. Plant material is an example of *biomass* (biological mass) which is a good renewable energy source. Biomass is material which is living or has been living.

Plants are able to trap the sun's energy using the chlorophyll in their leaves. They use the energy to convert carbon dioxide and water to carbohydrate and oxygen.

- 1 What is this process of energy conversion called?

Carbohydrates store energy. When they are burned in air, this energy is released. Wood has been a traditional fuel for many cultures. It is renewable, and is easily stored and transported. It can be converted to charcoal (carbon) by heating in the absence of air. Although burning wood releases carbon dioxide into the air, this is absorbed by the next generation of trees. Trees take a long time to grow and forests take up a considerable area of land which could otherwise be used for housing or farming.

Ethanol has the formula $\text{CH}_3\text{CH}_2\text{OH}$ and is an alcohol (alkanol). In Brazil, many cars (approximately 20%) run on ethanol rather than petrol. Sugar cane is a fast-growing tropical plant. The sugar (sucrose) it provides can be fermented to make ethanol. The fermented mixture is distilled to give pure ethanol. Ethanol is an excellent liquid fuel that can be used instead of petrol in cars (with slight modifications to the engine) or can be mixed with petrol. When ethanol burns it produces carbon dioxide and water. Ethanol itself is a clean, renewable fuel, but it needs a very large area to grow sugar cane, and the processes of refining sugar and ethanol do cause pollution.

Many plants produce vegetable oils — sunflower oil and peanut oil are examples. These oils could be used as fuels in the future. Scientists are trying to find ways of making diesel engines run efficiently using vegetable oils.

When vegetable and animal matter rots in the absence of air, *biogas* is given off. This is *anaerobic respiration*. The gas is usually about 60% methane and the rest is mostly carbon dioxide. It is a good fuel, particularly for cooking, heating and lighting the home. Biogas is particularly easy to make on farms. Manure and waste vegetable matter (such as straw) are allowed to rot in a closed tank. The solid waste remaining after biogas has been collected makes an excellent fertiliser. There are over 7 million biogas digesters in China, supplying the energy for 35 million people.

- 2 Explain what is meant by 'biomass' and give an example.
- 3 What chemicals are produced when ethanol burns?
- 4 Could Australia produce ethanol in a similar way to Brazil? What do you think would be the biggest drawback?
- 5 What is 'biogas'?
- 6 How is biogas made?
- 7 Give two uses for biogas.
- 8 Give two advantages and two disadvantages of using wood as a fuel.
- 9 Charcoal is mainly made up of what element?
- 10 What is the common name for 'sucrose'?



RESEARCH

- 11 Find out how fermentation of sugar can be used to make ethanol.
- 12 Define 'carbohydrate'.

Fuels

Fuels: substances that produce heat or power during combustion.

A fuel is a substance which gives out heat energy when it burns. Most fuels need to combine with oxygen (O) in order to burn to release their energy. Another name for burning is *combustion*. When carbon-based fuels burn they give out carbon dioxide (CO₂) and water (H₂O). Reactions giving out heat are called *exothermic* reactions.

Many of the fuels we use are *fossil fuels*. They are the remains of organic (living) matter from millions of years ago. They consist mainly of hydrocarbons — compounds containing the elements carbon (C) and hydrogen (H) only. Coal, oil and natural gas (methane, CH₄) are the major fossil fuels.

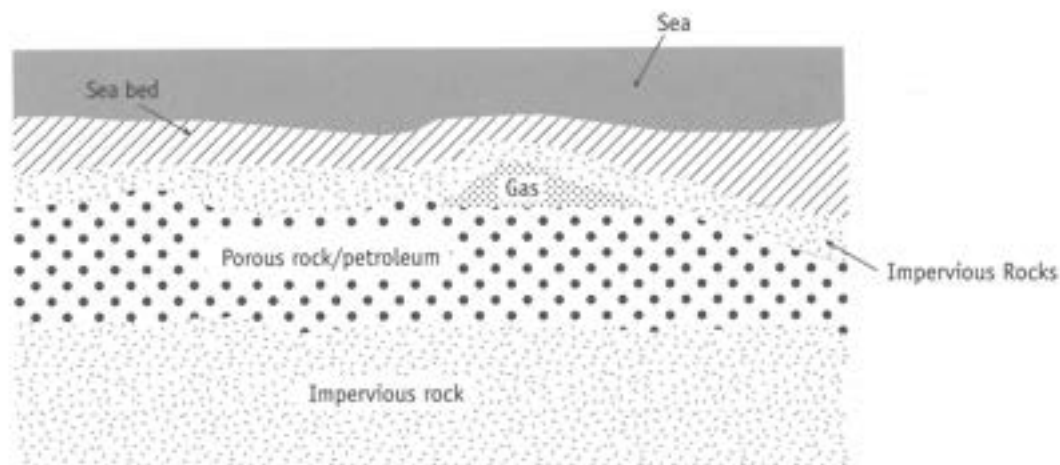
Coal is the remains of ferns, trees and other plants that lived about 300 million years ago, when most of the Earth was covered in swamps or forests. These plants stored energy from the sun in making their food via the process of photosynthesis. When trees died, they fell and gradually became covered by layers of vegetation. As time went on, the layers were crushed into the Earth, often by earthquakes, volcanic eruptions or layers of mud and sediment.

The dead trees were squashed by the layers above. This pressure and heat from inside the Earth changed the trees which eventually turned into seams of coal. Much of this coal lies deep underground, covered by layers of sediment. To mine these underground deposits of coal, deep shafts must be dug. Some coal seams have been pushed to the surface by earth movements. This coal can be mined by *open-cast* mining, once the surface rock has been bulldozed away.

Crude oil (petroleum) is thought to have been formed in a similar manner to coal. Millions of years ago, life flourished in the warm, shallow seas and oceans of the time. As these marine creatures died, their bodies fell to the bottom of the sea. Gradually the remains became covered with mud and silt. Decay (rotting), heat and pressure turned this organic (living) matter into oil.

Crude oil is often found under pressure, trapped in porous (spongy) rock between layers of impervious (impenetrable) rock. The porous rock is absorbent and can hold large quantities of oil. The impervious rock stops the oil from escaping.

Natural gas is often also present. This tends to collect above oil layers. The gas is under pressure. This means that, when a drill penetrates the oil-bearing layers, oil is forced out. The oil can be quite hot — up to 80°C. The first oil wells were dug around 1859 in the USA.



Fuels

Crude oil is a dark, viscous mixture. 'Viscous' means that it is not 'runny', or does not flow or pour easily. Crude oil is a mixture of hundreds of different hydrocarbons. Before the crude oil can be used, it must be refined. This process is called *fractional distillation*. The crude oil is heated strongly, turned into a vapour and distilled. The parts which boil (and condense) at different temperatures (*fractions*) are separated and collected. Each fraction has a different property and use.

Coal and oil take so long to form (millions of years) that they are described as *non-renewable* fuels. In other words, once they are used up no more will be made.

Methane is made whenever vegetation rots or decomposes. This happens in marshes, rice paddy-fields, sewage works, coal mines and the stomachs of cows (and humans).

Wood has been a widely-used fuel for thousands of years. It is probably the most widely distributed fuel in the world. We are currently using material formed from trees and forest vegetation that grew hundreds of millions of years ago for our fuel supplies.

- 1 What is a *hydrocarbon*?
- 2 Define a 'fuel'.
- 3 Give three examples of fuels.
- 4 What is meant by 'combustion'?
- 5 What is another name for petroleum?
- 6 From what is petroleum thought to have been formed?
- 7 Is petroleum a pure substance?
- 8 What is another name for natural gas?
- 9 Name two sources of natural gas.
- 10 Open-cast mining, because it is near the surface, can be cheaper than digging shafts to mine seams which are deep underground. Why might open-cast mining not always be popular?
- 11 Give the formula for methane.

State whether each of the following statements is true or false.

- 12 Exothermic reactions give out heat. _____
- 13 Coal and oil can be formed over a few hundred years. _____
- 14 A viscous mixture pours easily. _____
- 15 'Marine' creatures are found in the sea. _____



RESEARCH

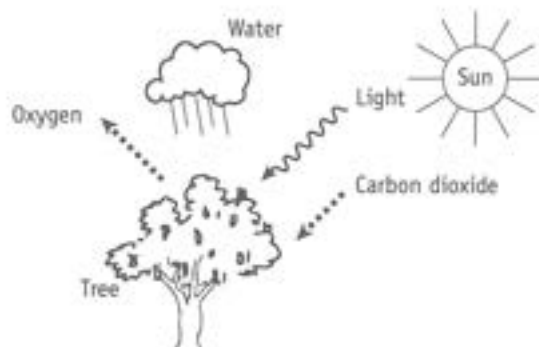
- 16 Find out how crude oil is turned into useful products in an oil refinery.
- 17 Give the locations of the major sites in Australia which produce:
 - natural gas
 - petroleum.
- 18 What was the name of the geological period when fossil fuels were formed?

Energy Transfer

Energy can be described as 'the ability to do work'. It can take several forms, such as sound, light, movement and heat. The unit of energy is the joule, which has the symbol J. The joule is very small so we tend to use kJ (kilojoules) or MJ (megajoules). There are a thousand joules in a kilojoule and a million joules in a megajoule. The average adult will need in the region of 5 MJ of energy a day to maintain health. A typical power drill or hair drier may be rated at 500 watts. This means that it uses 500 joules per second.

Consider a wood barbecue. When wood burns it is reacting with the oxygen in the air. This is an example of combustion. Carbon dioxide (CO_2) is one of the products of this reaction. The reaction heats up the surroundings, so heat is given out. This is an *exothermic* reaction. 'Exo' comes from the Greek and means 'out'. Heat energy is *given out* by the reacting system.

Trees take in energy from the sun. They grow by a chemical change called *photosynthesis*. Carbon dioxide from the air combines with water to make new substances. This is an *endothermic* reaction. 'Endo' also comes from the Greek and means 'in'. Energy is *taken in* by the system.



Green plants are producers — they make their own food. They contain chlorophyll and thus can *photosynthesise*. This process produces a food called glucose (a type of sugar) and also oxygen.

This process uses the energy of the sun (solar energy) to drive the reaction. This solar energy is converted to stored chemical energy in plants. This energy is used for growth by the plant and to provide energy for the plant during darkness when photosynthesis cannot take place.

Without organisms which photosynthesise (such as green plants) and organisms which live in water (such as phytoplankton), there would be no animal life. Animals would have no food to eat and no oxygen to breathe. Animals are consumers. If they eat plants, they are first-order consumers or *herbivores*. The chemical energy in the green plants is transferred to an animal when it eats the plant.

Glucose has the formula $\text{C}_6\text{H}_{12}\text{O}_6$. Each molecule contains six carbon atoms.

- 1 How many hydrogen atoms are present per molecule?
- 2 How many oxygen atoms are present per molecule?

Glucose is a carbohydrate. This means that:

- it contains carbon, hydrogen and oxygen
- hydrogen and oxygen are present in the ratio of H:O of 2:1 (this is the same as for water, H_2O).

The production of high-energy carbohydrates is an important step in storing the sun's energy in a form that can be used by animals as well as by plants. Carbohydrates are a major part in the diet of humans. *Respiration* is the process that animals use to break down food for energy. Large molecules are broken down into smaller ones. Carbon dioxide and water are produced, as well as heat energy, in this process.

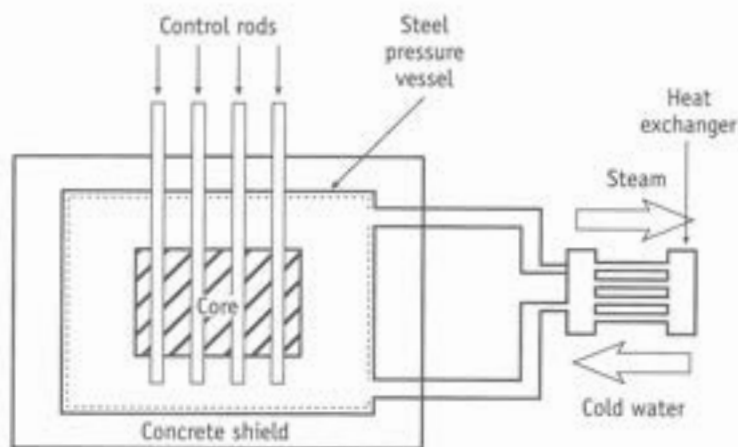
- 3 Define 'energy'.
- 4 What is meant by an *exothermic* reaction?
- 5 What is *solar* energy?
- 6 Write in words and then as a balanced equation the process of photosynthesis.
- 7 Sucrose has the formula $\text{C}_{12}\text{H}_{22}\text{O}_{11}$. Is it a carbohydrate? Give a reason for your answer.
- 8 Name the green substance present in plants which allows them to photosynthesise.
- 9 What would happen to animal life if there were no green plants?
- 10 Define 'respiration'.

Nuclear Energy

1 Before reading this sheet, write two or three lines about *nuclear power*.

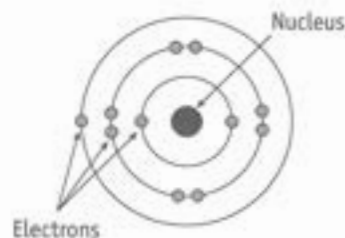
Nuclear power stations produce electricity in much the same way as 'conventional' coal and oil-fired power stations. Heat is used to turn water into steam. This steam is used to rotate turbines, which in turn drive generators to generate electricity. The difference is in *how* the heat is produced. Instead of burning fossil fuels, the energy is produced by *nuclear fission*.

Nuclear fission involves the splitting of atomic nuclei (centres of atoms) with the release of large amounts of heat and other forms of energy. This occurs in the nuclear reactor in a nuclear power station. The core contains fuel elements (uranium) and moderators (graphite). Moderators (control rods) control the chain reaction. The heat generated is used to turn water into steam to drive turbines and generate electricity. The reactor is surrounded by a steel pressure vessel and is enclosed by a thick concrete shield.



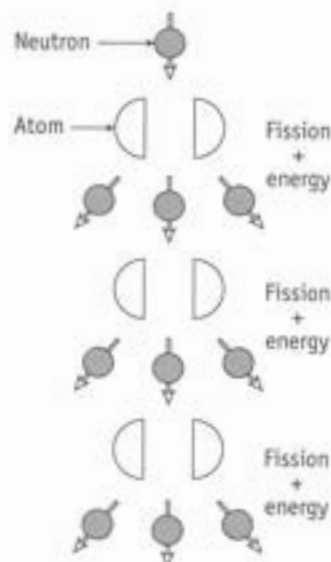
Some ships and submarines are powered by nuclear reactors. There are also some small nuclear reactors which produce only a small amount of electricity, but are mainly used for research and production of radioactive materials (radioisotopes) used in medicine. Lucas Heights nuclear reactor in Sydney is an example.

Atoms are made up of a very small, dense *nucleus* at the centre, surrounded by *electrons* moving very quickly around it. The nucleus is made up of *protons* and *neutrons*. Electrons have a negative charge, protons have a positive charge, and neutrons have no electrical charge.



In some (relatively very large) atoms the nucleus is unstable, which means that it can split easily into two smaller (more stable) nuclei. These new nuclei are not of the same element as the original nucleus. Having been formed by the break-up of a larger nucleus they no longer have the same number of protons and neutrons as the parent atom. Splitting is even more likely to happen if another particle (for instance a neutron) collides with this unstable atom. When a neutron hits a uranium atom, the atom can split into two smaller (different) atoms and eject two fast moving neutrons. These three neutrons may immediately collide with three more uranium atoms and split them. Each of these will also eject more neutrons which may collide with other uranium atoms, and so on.

This fast-growing reaction is called a *chain reaction*. When this process takes place, much heat energy is given out. In most nuclear power stations the fuel undergoing fission is *Uranium-235*.



Nuclear Energy

The energy released is due to mass being converted to energy. Note the famous equation shown below. This means that the energy released in a nuclear reaction is equal to the amount of mass converted multiplied by the square of the speed of light.

$$E = mc^2$$

The Uranium-235 nucleus undergoes fission when bombarded with neutrons inside the reactor core of a nuclear power station. The amount of energy released by fission is enormous. One kilogram of Uranium-235 can give out nearly 19 million kilowatt-hours of heat. Compare this with coal, where burning the same amount (1 kg) will give out less than 2 kilowatt hours of heat.

Radioactive materials must be handled with great care. Many of them must be stored securely for thousands of years before the radioactivity drops to a safe level. Thick lead or concrete shielding is used to protect nuclear laboratory workers from the effects of radiation. Workers dealing with radioactive materials often use disposable clothes which are sealed in containers and stored after use. A type of 'badge' containing photographic film is worn to detect if a person has been exposed to radiation. Radioactivity 'fogs' film and this can be detected when the film is developed.

ARGUMENTS AGAINST NUCLEAR POWER

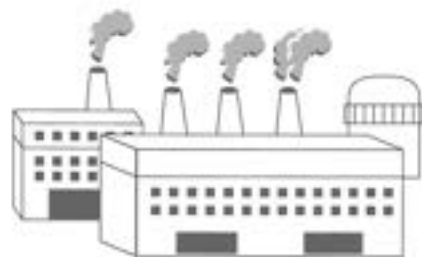
- Very dangerous rays and material could leak out if an accident occurred.
- Dangerous rays come from the fuel rods that have been used in reactors.
- 2 Give two other arguments *against* nuclear power.

ARGUMENTS FOR NUCLEAR POWER

- Nuclear power is very safe. There have been less accidents than in other kinds of power stations.
- People are well protected from radiation from reactors.
- 3 Give two other arguments *for* nuclear power.

Conventional (fossil fuel) power stations are not without their drawbacks. They use non-renewable resources (coal and oil), cause pollution of the atmosphere (acid rain) and emit carbon dioxide to increase the 'Greenhouse Effect' which contributes to global warming. About 95% of Australia's electricity is generated by fossil-fuel power stations — most using coal.

- 4 What particles make up the nucleus of an atom?
- 5 How is steam used in a nuclear reactor to make electricity?
- 6 What is used to stop radiation from affecting nuclear workers?
- 7 What means could be used to see if someone is being exposed to nuclear radiation?
- 8 In the equation $E = mc^2$, what do the symbols represent?
- 9 What happens to atoms in nuclear fission?
- 10 What type of energy is released in nuclear fission?
- 11 Compare the energy given out when 1 kilogram of coal is burned and when 1 kilogram of uranium undergoes fission?
- 12 Describe how a chain reaction can take place in Uranium-235.



RESEARCH

- 13 Find out how the Sun produces energy in a process called *fusion*. Describe any advantages and disadvantages this process might have compared to *fission* in nuclear power stations.
- 14 Uranium does not occur uncombined in nature. Two ores of uranium are *pitchblende* and *carnotite*. Give the chemical composition of these substances and explain how uranium is extracted from them.

F1 SCIENTIFIC METHOD I

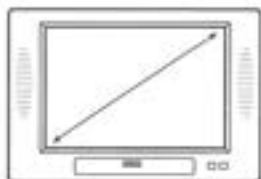
- 1 Hypothesis.
- 2 Observation.
- 3 Hypothesis.
- 4 Hypothesis.
- 5 Observation.
- 6 Qualitative (no numbers are used).
- 7 Weeks 3 and 4.
- 8 There might be a connection/correlation. Other factors may play a part. Further investigation is needed before they could make this statement.
- 9 A science using fossil remains to study the life of past geological periods.

F2 SCIENTIFIC METHOD II

- 1 Thermometer.
- 2 To measure the temperature of the liquid.
- 3 Beaker.
- 4 Measuring cylinder.
- 5 A beaker is not used to measure the volume of liquid accurately — a measuring cylinder is.
- 6 Volume of water, timing intervals, room temperature, initial temperature.
- 7 Time, remains consistent and cannot be changed.
- 8 Temperature.
- 9 Temperature varies with time elapsed.
- 10 80°C.
- 11 25°C.
- 12 The results for the white beakers.
- 13 The water in the silver containers cools more slowly than that in the red or black containers; or the water in the black containers cools more quickly than that in the red or silver containers.
- 14 It has a twin-walled container with a vacuum between those walls.

F3 TABLES

- 1 Supercool.
- 2 Technocool.
- 3 Arctic.
- 4 Freezefast.
- 5 Arctic, Supercool.
- 6 Arctic.
- 7 Koola, \$2.50; Technocool, \$2.00; Freezefast, \$1.59; Polar, \$1.69; Arctic, \$2.49; Supercool \$1.92.
- 8 Guarantee, star rating/energy efficiency, where it is made, design of interior, etc.
- 9 Six.
- 10 Three.
- 11 Newtel.
- 12 34 cm.
- 13 DigiTV.
- 14 \$820.
- 15 SupaTV.
- 16 No, you need to look at it for such features as picture/sound quality, ease of use of controls, etc.
- 17 Diagonally from corner to corner.

**F4 AND F5 THE PENDULUM**

- 1 50 g: Average = 12.60 s, Periodic time = 0.63 s
100 g: Average = 12.49 s, Periodic time = 0.62 s
150 g: Average = 12.63 s, Periodic time = 0.63 s
- 2 Teacher to check.
- 3 None, the times are the same (within experimental error).
- 4 50 cm: Average = 14.21 s, Periodic time = 0.71 s
75 cm: Average = 17.49 s, Periodic time = 0.87 s
100 cm: Average = 20.22 s, Periodic time = 1.01 s
- 5 Teacher to check.
- 6 As length increases, periodic time increases.
- 7 10 cm: Average = 14.46 s, Periodic time = 0.72 s
20 cm: Average = 14.47 s, Periodic time = 0.72 s
- 8 Teacher to check.
- 9 None, the times are the same (within experimental error).

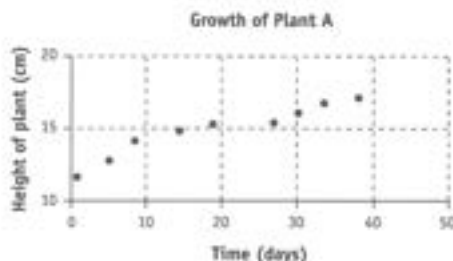
F6 SURVEYS

- 1 104.
- 2 Basketball = 12%, Australian football = 18%, netball = 14%, rugby union = 3%, rugby league = 8%, golf = 3%, soccer = 16%, lawn bowls = 3%, fishing = 13%, roller hockey = 2%, canoeing = 4% and baseball = 2%.

- 3 Teacher to check.
- 4 Netball.
- 5 2.
- 6 No reply.
- 7 Don't = 45, Seldom = 33, Regularly = 30, No reply = 12.
- 8 55%.
- 9 40%.
- 10 5%.
- 11 It was not a representative sample of people questioned, or not a large enough sample, people surveyed may have bought beer from other outlets, or may not have given a truthful response.
- 12 Brewers, firms who sell beer, Government health organisations, etc.
- 13 Take more samples, use different times, use different places, ask different questions, etc.
- 14 Advantages: quick, simple. Disadvantages: results only represent a particular group of people (those who read the paper, those who care about the topic and those who have a phone) and they can't comment on the topic.
- 15 For example, draw up a questionnaire, ask a large number of people in the target age group, same number males/females, etc.
- 16 By phone or with face-to-face questions. Accuracy is variable can be $\pm 3\%$.

F7 PLANT GROWTH

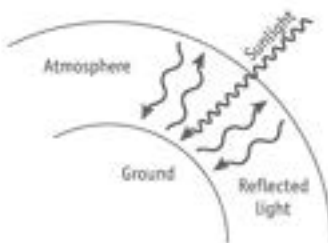
- 1 4.8 cm.
- 2 No mention of using 'Gro-betta'. They were asked to investigate the effect on growth of the liquid fertiliser.
- 3 Same temperature, lighting, same size of plant pot, amount of soil, etc.
- 4 No, there should have been experiments where no fertiliser was added to act as a 'control'.
- 5 See graph. 'Time' is the independent variable and 'height' is dependent variable.



- 6 Around 18 cm, by extrapolation of the graph or calculation based on figures in the table. Assume the trend of growth is consistent.

G1 AND G2 THE ATMOSPHERE AND GREENHOUSE EFFECT

- 1 See diagram.



- 2 Water H_2O , carbon dioxide CO_2 , methane CH_4 , nitrous oxide NO , ozone O_3 .
- 3 Can be clearly seen through.
- 4 No, temperature would vary too much.
- 5 Car heats up very quickly, to a very high temperature.
- 6 The Sun.
- 7 They might melt. Low lying countries can be submerged.
- 8 Heat.
- 9 Doesn't contain CFC, the propellant doesn't harm the ozone layer.
- 10 Use less fossil fuels, plant more trees.
- 11 Sunlight.
- 12 Not enough oxygen for life at that height.
- 13 Some areas will be more productive, while others will suffer.
- 14 Increase them.
- 15 Nitrogen (78%), oxygen (21%) and argon (just under 1%).
- 16 Troposphere: the lowest and densest part of the Earth's atmosphere. Stratosphere: layer from approximately 15 km to 50 km above Earth's surface. Mesosphere: beyond stratosphere, the coldest part of the atmosphere. Ionosphere: the layer that reflects radio waves, around 50 km to 450 km above Earth's surface.

- R17 For example hydrochlorofluorocarbons (HCFCs) and hydrofluorocarbons (HFCs) which do affect the ozone layer, but much less than CFCs. Gaseous hydrocarbons, which do contribute to global warming.
 R18 Decomposing vegetable matter in swamps or rice paddies.

G3 AND G4 POLLUTION

- 1 Elements: carbon (C), lead (Pb), copper (Cu), oxygen (ozone) (O_3), mercury (Hg).
 Compounds: sulfur dioxide (SO_2), carbon monoxide (CO).
- 2 Coal, oil, natural gas.
- 3 Smoke and fog combined.
- 4 O_3 , a form of oxygen which screens out UV rays.
- 5 Lead damages health.
- 6 Something (often made by humans) which contaminates an environment.
- 7 Acid rain.
- 8 To do with water.
- 9 The interaction of food chains ('who eats who') in an ecological community.
- 10 Waste liquids or other materials carried off by sewers.
- 11 Yes, lead damages health.
- 12 To replace nutrients removed/lacking, to make crops grow better.
- 13 To kill pests.
- 14 Herbicide kills weeds (plants) and an insecticide kills insects.
- R15 Large amounts of nutrients (for instance from fertilisers) in water cause rapid growth of algae and bacteria. This reduces the oxygen present for fish, etc.
- R16 Gaseous hydrocarbon is one example.
- R17 Toxic mercury compounds from industry found their way (via river systems) to the sea where they were taken up by fish. Fish didn't die but mercury was stored in their bodies to eventually be consumed by humans with devastating effects, documented by W. Eugene Smith.

G5 QUADRAT SAMPLING

- 1 Teacher to check.
- 2 Totals in 36m²: Daisies = 37, Dandelions = 60, Oxalis = 101. Would expect students' answers (for 10 m²) to approximate: Daisies = 10, Dandelions = 17, Oxalis = 28. In 1000 m² would expect students' answers to be in the region of: Daisies = 1027, Dandelions = 1667, Oxalis = 2806. (Note: these will be variable due to sampling variations.)
- 3 That the sampling technique is accurate, and the same ratio of plants exists in all of the field.
- 4 Much quicker.
- 5 Take more samples. Take samples over a larger region of the field.

G6 CLIMATE

- 1 August, 27°C.
- 2 January, -4°C.
- 3 August.
- 4 February.
- 5 566 mm.
- 6 47.2 mm.
- 8 Northern, because winter is Dec/Jan/Feb and summer is Jul/Aug/Sep.
- 9 Weather is how hot, cold, cloudy, windy, etc it is at particular time, while climate is a pattern of weather over a period of years.

H1 THE EARTH'S RESOURCES

- 1 Air.
- 2 Once used up, there will be no more.
- 3 Increased population, greater use of technology, improved standard of living etc.
- 4 Houses, fences, fuel, etc.
- 5 Roads, sculptures, monuments, etc.
- 6 Drinking, washing, recreation, etc.
- 7 Artificial, made by humans.
- 8 The part of the Earth where life can exist.
- 9 Lithosphere: crust. Hydrosphere: where water is present. Atmosphere: gas around the Earth.
- 10 Fuel.
- R11 Natural source of supply.
- R12 Paint, varnish, creosote, copper-based compounds, etc.
- R13 It is dried out slowly.
- R14 A compound containing many repeating chemical units, for example.
- R15 Some animals have special microorganisms in their digestive tracts.

H2 BUILDING MATERIALS

- 1 It is available, strong, durable, etc.
- 2 It is difficult to cut/quarry.
- 3 Able to stand up well to rain, wind, and extremes of temperature.
- 4 Long-lasting.
- 5 Monuments, sculptures. Its attractive appearance (lustre), and ability to be shaped.
- 6 It is strong, durable, cheap, and can be poured into shapes.
- 7 Materials were present, the technique was used in the settlers' home country, it is relatively easy to do.
- 8 It is a renewable resource, easy to work with, and has an attractive appearance.
- 9 It lasts for a long time, has an attractive appearance, and is strong.
- 10 It lasts for a long time, comes in handy-sized units, and is strong.

- 11 Steel-mesh, corrugated iron, bushes, etc.
- R12 Limestone.
- R13 Around 1820 in England, where lime and clay were heated strongly.
- R14 Building material made from sun-dried clay/soil.

H3 RESERVES AND RESOURCES

- 1 Oil, water.
- 2 We estimate that there is 360 000 000 tonnes of copper in the Earth available to mine.
- 3 No more will be discovered, and the amount used will not vary.
- 4 8 000 000 tonnes per year.
- 5 The amount of the substance you want that it contains.
- 6 Discovery of new reserves, finding alternatives.
- 7 & 8 Teacher to check.
- 9 Recycle, use alternatives.
- R10 Check 'commodities' section of national newspaper.
- R11 Proportion of, for example, an ore body that can be extracted economically.
- R12 The potential wealth of a country, for instance.

H4 ROCKS AND MINERALS

- 1 A rock has variable composition, a mineral has a fixed composition and physical properties.
- 2 Silicon (Si), oxygen (O), sulfur (S), carbon (C), lead (Pb), magnesium (Mg), zinc (Zn), aluminium (Al), iron (Fe).
- 3 To help identify them (there are over 3000 types).
- 4 Silicates, non-metallic, metallic and gems.
- 5 Haematite, sphalerite, bauxite respectively.
- 6 Quartz, feldspar and mica.
- 7 Diamond.
- 8 Colour, shape, cleavage, lustre, streaking and hardness.
- R9 They come from living sources.
- R10 This scale arranges ten common minerals in order of increasing hardness and gives them a number: 1, talc; 2, gypsum; 3, calcite; 4, fluorite; 5, apatite; 6, feldspar; 7, quartz; 8, topaz; 9, corundum; and 10, diamond.
- R11 Lubricant, pencils, electrodes, etc.
- R12 Naturally occurring homogeneous substance.
- R13 Opals.

H5 ORES AND METALS

- 1 Iron and oxygen.
- 2 Aluminium and oxygen.
- 3 The amount of ore in the ground that we know about and that is worth mining.
- 4 297 500 tonnes Fe and 106 250 tonnes Al.
- 5 Approximately 50 years from now.
- 6 New reserves may be found, less may be used, more recycled, etc.
- 7 More may be used, estimates of the ore in the ground may be too optimistic, etc.
- 8 Gold is much more scarce and more valuable than iron.
- 9 How much there is of it (abundance), demand for it and how difficult it is to extract.
- 10 Gold (Au), silver (Ag), copper (Cu), iron (Fe), aluminium (Al).

H6 IRON AND STEEL

- 1 Uncombined with other elements.
- 2 Iron pyrites (FeS_2), haematite (Fe_2O_3), magnetite (Fe_3O_4).
- 3 $CaCO_3$.
- 4 CO_2 .
- 5 CO.
- 6 Heat resistant.
- 7 Once it is used up, there will be no more.
- 8 Iron ore, coke, limestone and air.
- 9 Impurities produced in making iron in the blast furnace.
- 10 No, it is too brittle — it won't bend well. Likely to collapse under the weight and vibration of traffic.
- 11 It rusts (combines with oxygen and water).
- 12 It is cheap, easily shaped and relatively strong.
- 13 Proximity to raw materials, to transport, to labour and to markets.
- R14 When cultures used iron to make implements, for example, tools and weapons. It happened at different times in different places.

H7 ALUMINIUM

- 1 Hydrogen (H), chlorine (Cl), sodium (Na), gold (Au), copper (Cu), silver (Ag), aluminium (Al), oxygen (O), carbon (C), magnesium (Mg).
- 2 Aluminium oxide.
- 3 Breaking down a liquid substance by passing electricity through it.
- 5 Aluminium and oxygen.
- 6 It melts, flows to the bottom (more dense) and can be drained off.
- 7 Large quantities of electricity are needed.
- 8 It has a high strength to weight ratio, which keeps weight down without compromising the strength of the metal.
- 9 It is lighter and cheaper.
- 10 Overhead cables.
- 11 8%.
- 12 Thermal conductivity, flexibility, it doesn't react with food, etc.
- R13 Aluminium is too reactive.
- R14 Gove (NT), Mitchell Plateau (WA), Weipa (Qld), etc.

H8 GLASS

- 1 Sand (silicon dioxide), sodium carbonate and calcium carbonate (limestone).
- 2 They are lighter, and less likely to shatter.
- 3 They scratch more easily, and their optical properties not as good.
- 4 Bend light.
- 5 It is strong and resists heat and chemicals.
- 6 Over 4000 years.
- 7 It is light and strong.
- 8 Fibre optic cables carry more information than equivalent sized copper cables.
- 9 Glass tubes/rods, fibreglass (insulation), ornaments, etc.
- R10 Transparent — can be seen through clearly. Translucent — transmits light but diffuses it so objects can be seen, but not clearly. Opaque — blocks light, cannot be seen through.
- R11 Consisting of glass.

H9 PAPER

- 1 Nearly 2000 years.
- 2 Clay, parchment, vellum, papyrus.
- 3 Waste.
- 4 Wood.
- 5 Teacher to check.
- 6 Wrapping/packaging, filters, stamps, toilet paper, etc.
- 7 It might not be biodegradable, and might harm environment.
- 8 Reduces waste, and saves trees.
- 9 13 600 m³
- R10 They have been found to be toxic, carcinogenic, etc.
- R11 If mixed with water, heated, broken down to a thick paste, the ink is removed, then it is mixed with fresh wood pulp and made into paper.

H10 MINING AND MINERALS:**Assignment**

- 1 Teacher to check.
- 2 Teacher to check.
- 3 Broken Hill: silver, zinc, tin and lead. Mount Isa: copper, zinc, lead. Kalgoorlie: gold, nickel. Tennant Creek: copper, gold. King Island: tungsten, titanium.
- 4 Au.
- 5 Coins, jewellery, electronics, dentistry, etc.
- 6 Kalgoorlie (WA), Patterson Ranges (WA), Tennant Creek (NT), Mount Morgan (Qld), etc.
- 7 Teacher to check.
- 8 Opals.
- 9 SA.
- 10 A mineral containing a worthwhile amount of a substance such as metal.
- 11 Aluminium.
- 12 Iron.

I1 BALANCE IN A NATURE

- 1 Change in climate, change in habitat.
- 2 Organism = living creature, beneficial = helpful, pathogenic = disease causing, dung = excrement, larvae = young insects, breed = produce offspring, pasture = grass, native = indigenous.
- 3 Reduce amount of dung, reduce numbers of flies, replace existing insects.
- 4 $10 \times 500 = 5000 \text{ cm}^2$ or 0.5 m^2 .
- 5 $40\,000\,000 \times 0.5 = 20\,000\,000 \text{ m}^2$ or 2000 hectares.
- 6 730 000 hectares.
- 7 They lay eggs in dung and the larvae eat the dung when they hatch, further breaking it down by digesting it before releasing it as waste.
- 8 Other insects, invertebrates, weathering.
- 9 No, they were not found here until introduced by early European settlers.
- 10 Members of the same species present in an area.
- 11 A space containing organisms interacting with each other and their surroundings.
- R12 Liquid medicine administered to an animal.

I2 AND I3 COMPONENTS OF ECOSYSTEMS

- 1 Water availability (for example rainfall), sunlight, ground (for example soil/sand/rocks), etc.
- 2 Desert, seashore, billabong, dry outback, coral reef, etc.
- 3 A place to live and breed (and food).
- 4 All life would eventually cease.
- 5 Cow receives $3 \div 1000 = 0.3\%$
- 6 Study of living things and their environment.
- 7 Members of different species living together.
- 8 A producer.
- 9 Species.
- 10 Consumer.
- R11 A type of ecological region, for example, tundra, rain forest, grassland, or desert.
- R12 Around 4% of body mass for an adult animal, or $0.04 \times 600 \text{ kg} = 24 \text{ kg/day}$.
- 13 'Is eaten by'
- 14 To do with the sea.
- 15 Plankton.
- 16 Humans.

- 17 They would increase (assuming nothing else eats them).
- 18 They would decrease (unless they can find and adapt to replacement food sources).
- 19 They would increase (assuming nothing else eats them).
- 20 No, often organisms are eaten by more than one predator, etc.
- 21 It would break down, because components are not returned to the system.
- 22 Breeding rate, food, shelter.
- 23 Consumers eat other organisms (plant or animal), carnivores specifically eat other animals.
- R24 Animals living in seas/oceans.
- R25 Some can emit a blue/green light, they are found from near the surface to depths of 2000 m, they eat diatoms, etc.
- R26 Plankton is divided into zooplankton and phytoplankton, it is found in fresh and salt water, it can breed rapidly causing colouration of water, etc.
- R27 Toothed whales eat fish, etc. Baleen whales (such as the blue whale) strain seawater and eat krill and plankton. Killer whales are carnivores.

I4 FOOD PYRAMIDS AND ENERGY

- 1 The energy was used by the organism for metabolic processes to keep it alive.
- 3 From bottom: vegetation, herbivore, carnivore, secondary carnivore.
- 4 Predators need large numbers of prey. Ratio of predator to prey is large.
- 6 From the bottom: producer, first consumer, second consumer, tertiary consumer.
- 7 100%, 10%, 1%, 0.1%.
- 8 They get larger.
- 9 The amount of energy transferred per link means very little is left at the end. The size of the predator would be very large or a very large prey population would be needed.
- 10 Initial population, rate of reproduction, food supply, presence of predators, etc.
- 11 Trophic (nutritional) level.
- 12 Ecosystem.
- R13 Concentration of a (harmful) substance increases as it passes up a food chain. (See DDT).

I5 INTERACTION BETWEEN ORGANISMS I

- 1 It spreads the workload, and increases the chances of survival.
- 2 It provides more chance of mating, confuses predators (harder to pick out), increases success in finding food, and increases ability to protect territory.
- 3 Photosynthesis.
- 4 No, only mice gain.
- 5 They carry disease.
- 6 Mosquito.
- R7 Mutualism: Legumes and Rhizobium bacteria. Emus and nitre bush. Humans and pets. Commensalism: Ferns (epiphytes) growing on trees. Ants and aphids. Colon bacilli and plant eating animals. Parasitism: Animal host and tapeworms, ticks, viruses, and bacteria.

I6 INTERACTION BETWEEN ORGANISMS II

- 1 Interdependence of two different species.
- 2 Mutualism — both benefit. (Wildebeest are rid of insects. Birds gain food.)
- 3 Mutualism — both benefit. (Remora gain food. Sharks have skin cleaned.)
- 4 Parasitism — tapeworm gets food but damages host.
- 5 Mutualism — both benefit. (Plants are pollinated and bees gain food.)
- 6 Mutualism — both benefit. (Plants gain nitrogen and bacteria gain food.)
- 7 Commensalism — ferns gain support, while trees gain nothing.
- 8 Mutualism — both benefit. (Carob seeds are made viable and spread, and animal gains some nutrition from the fruit.)
- 9 This can be viewed in two ways. Birds and mistletoe is an example of mutualism. (Birds gain food from fruit and mistletoe seeds are spread.) Mistletoe and trees is an example of parasitism. (Mistletoe takes nourishment from tree and, presumably, harms it.)
- R10 Grows to 15 m in height, fruit is said to resemble chocolate in taste, grows in warm climate, it is a legume, etc.
- R11 Another name for a wildebeest.
- R12 They are jawless fishes (resembling eels), they suck blood from other fish, they exist in fresh and salt water, they do not have a bony skeleton, etc.

I7 A POPULATION EXPLOSION

- 1 Teacher to check.
- 2 Population is the dependent variable, time is the independent variable.
- 3 200 million (approximately).
- 4 400 million (approximately).
- 5 1800 million (approximately).
- 6 Teacher to check.
- 7 In excess of 7000 million. The slope is exponential, answers will vary considerably.
- 8 That the population continues to grow at same rate.
- 9 2000 years (approximately).
- 10 1650 years.
- 11 40 years (approximately).
- 12 Present time.
- 13 No, resources and food will run out.
- 14 We can alter our environment, create new food sources, and exercise control of populations.

18 PEST CONTROL

- 1 Pigeon, duck, etc.
- 2 Locust, beetle, many larvae, etc.
- 3 Kills insects.
- 4 For every \$1 spent on insecticide, farmers gain \$4 in increased production.
- 5 May harm farmers using them, may reach waterways, and may stay in system (not degrade).
- 6 As a physical barrier, it stops sunlight reaching the weeds so photosynthesis cannot take place.
- 7 Pesticides kill a wide range of organisms, biological controls can attack only one species.
- 8 The surviving rabbits have developed a resistance/immunity to disease.
- 9 They damage the environment, displace other (native) animals, and eat pasture intended for farm animals.
- 10 Fencing, trapping, shooting, digging up warrens, shooting, use ferrets, etc.
- R11 To eat grey-back beetles which were destroying sugar cane fields in Queensland (1935).
- R12 Examples where diseases or parasites of the target insect are used, or the breeding of plants to be resistant to pests.
- R13 Dichloro diphenyl trichloroethane. It is a very efficient insecticide. It enters food chains and by 'biomagnification' causes problems with animals eating insects sprayed with DDT. Some insects do become immune.

19 ECOLOGY: CLOZE

- | | | |
|------------------|----------------|-----------------|
| 1 organisms | 11 consumers | 21 habitat |
| 2 sun | 12 herbivores | 22 population |
| 3 solar | 13 carnivores | 23 community |
| 4 photosynthesis | 14 green | 24 ecosystem |
| 5 producers | 15 decay | 25 temperature |
| 6 plants | 16 decomposers | 26 symbiosis |
| 7 eaten | 17 web | 27 mutualism |
| 8 incorporated | 18 system | 28 relationship |
| 9 food | 19 environment | 29 parasitism |
| 10 warm | 20 ecology | |

J1 ENERGY I

- 1 For example: 'the cause of change', 'makes things happen', 'does something', etc.
- 2 Chemical energy.
- 3 Positional energy.
- 4 Chemical energy.
- 5 Chemical energy.
- 6 Elastic energy.
- 7 No, we only see effects, or what it has caused to happen.
- 8 No (it has no mass).
- 9 No.
- 10 No. Some energy is used in body maintenance by a cyclist. Some energy supplied to the bicycle is turned to heat and sound as well as movement.
- 11 Batteries make it portable, it can be used at times of power failure or when there is no access to mains electricity.
- 12 It is more expensive to run, the device can't be one which uses large currents.
- 13 Anything which is moving.
- R14 Energy = (mass) $\times$ (speed of light squared). This has been used in atomic bombs, converting mass to energy.
- R15 A chemical reaction occurs which involves a flow of electrons; or electric current.
- R16 The wire heats up and glows when the current flows. See, 'Resistance' in Book 1 of this series.

J2 ENERGY II

- 1 Chemical energy $\rightarrow$ heat energy and light energy.
- 2 Movement energy $\rightarrow$ sound energy.
- 3 If it is a 'clockwork' clock, elastic (potential) energy $\rightarrow$ kinetic energy $\rightarrow$ sound energy. If it is an electric clock, electrical energy $\rightarrow$ kinetic energy $\rightarrow$ sound energy (If it is a battery-powered, could include chemical energy at the first stage).
- 4 Electrical energy $\rightarrow$ kinetic energy (plus heat energy and sound energy).
- 5 Electrical energy $\rightarrow$ kinetic energy $\rightarrow$ sound energy (If the clock is battery-powered, could include chemical energy for the first stage).
- 6 No, energy can neither be created nor destroyed, only changed to another form ('Law of conservation of energy').
- 7 Any device that uses a battery and has movement, for example a cordless drill, a personal stereo, etc.
- 8 1000 joules.
- 9 They could be on a diet.
- 10 Coal, wood, paraffin, petrol, diesel, etc.
- R11 One newton (N).

J3 ENERGY FROM BIOMASS

- 1 Photosynthesis.
- 2 Biological mass — anything which is living or has been living.

- 3 Carbon dioxide and water.
- 4 Yes (sugar cane in Queensland). Amount of space available, the amount of water needed to grow enough sugar.
- 5 Methane (60% approximately) and carbon dioxide.
- 6 Vegetable and animal matter rotting in the absence of air.
- 7 Cooking, heating and lighting.
- 8 Advantages: It is renewable, easily stored and easily transported. Disadvantages: it takes a long time to grow and it takes up a lot of space.
- 9 Carbon.
- 10 Sugar.
- R11 Fermentation (brewing). Yeast is added to a solution of sugar and kept at a steady temperature, for example 25°C. Ethanol (alcohol) and carbon dioxide are produced.
- R12 A compound containing carbon, hydrogen and oxygen with hydrogen and oxygen atoms in ratio 2 : 1.

J4 AND J5 FUELS

- 1 A substance containing only hydrogen and carbon.
- 2 Something which gives out heat and light when it burns.
- 3 Coal, wood, methane, LPG (liquefied petroleum gas), paraffin, diesel, etc.
- 4 Burning, when a substance combines with oxygen releasing heat, light, etc.
- 5 Crude oil.
- 6 Marine (sea) organisms.
- 7 No, it is a mixture of hydrocarbons.
- 8 Methane.
- 9 Oil fields, coal mines, marshes, paddy fields, sewage works, cows (ruminants).
- 10 It causes damage to environment, and doesn't look good.
- 11 CH_4 .
- 12 True.
- 13 False.
- 14 False.
- 15 True.
- R16 Fractional distillation/refining is used. The products include LPG, petrol, paraffin (kerosene), diesel, lubricating oil, fuel oil, bitumen, etc.
- R17 • Cooper Basin, Bass Strait, Scott Reef, etc.
• Bass Strait, Dampier.
- R18 Carboniferous.

J6 ENERGY TRANSFER

- 1 Twelve.
- 2 Six.
- 3 The ability to do work.
- 4 One that gives out heat.
- 5 Energy from the sun.
- 6 Carbon dioxide plus water gives glucose (a sugar) plus oxygen.
 $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$
- 7 Yes, it contains carbon. Hydrogen and oxygen are present in the ratio of 22 : 11, or 2 : 1.
- 8 Chlorophyll.
- 9 Animal life would die out.
- 10 The process by which animals break down food to provide energy. (Note: 'breathing' is not acceptable in this context.)

J7 AND J8 NUCLEAR ENERGY

- 1 This is a discussion point, so there are no 'right answers' only opinions. There might be mentioned 'too dangerous', 'only choice to avoid global warming', etc.
- 2 The waste remains dangerous for many years, terrorists could sabotage plants, we don't know how to handle the waste, the dangers of transporting the fuel, etc.
- 3 It saves fossil fuels (coal, petroleum), it doesn't contribute to CO_2 or global warming, the volume of waste is small, there are large deposits of uranium in Australia.
- 4 Proton and neutron.
- 5 Shielding (lead or concrete) and disposable clothes.
- 6 Steam drives turbines which generate electricity.
- 7 Using a 'badge' containing photographic film.
- 8 E is energy, m is mass and c is the speed of light.
- 9 They are split.
- 10 Nuclear energy, mainly heat energy.
- 11 1 kg of coal produces 2 kW hours of heat (approximately) and 1 kg of uranium provides 19 000 000 kW hours of heat (approximately).
- 12 A fast moving neutron hits a nucleus and this is split. The neutrons released split more nuclei, which release more neutrons, and so on.
- R13 Hydrogen nuclei join (fuse) to form heavier helium atoms, releasing energy. No radioactive waste is produced, but conditions are very difficult, such as very high temperatures and pressures, etc.
- R14 Pitchblende (uranite) is impure uranium oxide, UO_2 , (also contains thorium, lead and rare earths). Carnotite contains uranium, radium, potassium, vanadium and oxygen. Extraction methods include ion exchange, solvent extraction and using volatility.



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